

POSTHUMANISM

WEEK 1

Posthumanism Context 1: Technologies

It is a school of thought and critical artistic practice which argues for the radical decentering of the traditional, sovereign, coherent, and autonomous human in order to demonstrate how humans have always evolved with, been constituted by, and been part of multiple forms of life and machines. It views the human as entangled with non-human life forms and the non-living in a mutually dependent relationship while striving to demolish hierarchies such as human/non-human, living/non-living.

Domains where technology increasingly interferes with, interfaces with, alters, and manages human life systems and ecosystems:

- Artificial intelligence
- Life sciences, especially genetics, medicine, and xenotransplantation
- Consumer technologies
- Neural and cognitive sciences

New information and digital technology offer the promise of escaping the material body into another medium. It also produces a networked self, where humans and machines are integrated, and we become our self-tracking devices. We create a database-mediated self where humans are rendered into a matrix of numbers. Biotechnologies such as cloning blur the line between natural and artificial life. Genetic engineering and xenotransplantation blur the boundaries between species and bodies.

Enhancement Technology

- **Biological enhancement:** Genetic alterations to improve the next generation of humans with better immunity, physiology, and cognitive abilities.
- **Bionic enhancement:** The replacement of biological parts with synthetic parts that experience less wear and tear.
- **Virtual enhancement:** The digitization of memories and personality traits so that these can be downloaded into robotic substitutes at a future date.

Hans Moravec, in *Mind Children: The Future of Robot and Human Intelligence* (1988), discusses the digitization of memory and the potential to upload consciousness into robots, granting a form of immortality. The creation of artificial beings, such as robots increasingly similar to humans, blurs the borders between sentient humans and sentient non-humans.

Smart devices, from phones to cars, now embody a form of intelligence by calculating, predicting, and responding to their environment. Artificial intelligence technology pushes machines toward surpassing human abilities. There is also a growing discussion around "moral bio-enhancement" to produce humans who are more compassionate, artistic, or even more violent. These enhancements are meant to create "better humans."

Humans with necessary prosthetics or implants seek to transcend the limitations of the human body, particularly weaknesses, diseases, aging, and mortality.

Posthumanist thought does not see technology or devices as additions to a pre-existing human. It views the human as always having used technology—the first technology being the human body itself. It argues that humans have always co-evolved with technology, from fire and weapons to contemporary digital devices.

There are two ways of thinking about the current technologically mediated person, whose life systems incorporate non-human life, devices, and elements like earth, air, and space, creating mutual dependencies and co-evolution.

1. These connections can be seen as deviations from the human.
2. These connections can be seen as the *truer* nature of the human.

In the first case, this view relies on the notion of a human distinct from technology, existing in a pure form, autonomous from devices, settings, and other non-human forms.

In the second case, it suggests that we are reaching a deeper understanding of the human, co-evolving with and dependent on other forms. This is posthumanism.

Context 2: Humanism and Its Critiques

Humanism traditionally posits the human as a self-aware subject marked by rational thinking and intelligence, capable of defining actions based on needs, desires, and wishes, and executing these to produce history. The human has been treated as male and universal, always in the singular as "the human," defined by features like rationality, authority, autonomy, and agency. Humanism positions this human as the center of the world.

Critics of humanism question the notion of the human as the center of the universe, the autonomous nature of the human mind, and the agency that supposedly arises from autonomy. The "universal human" is constructed through a process of exclusion, whereby certain ethnic, religious, or racial groups are categorized as "less than human," ranking humans above animals and asserting that nature exists to serve the male/human.

Traditional, Eurocentric humanism has historically positioned homosexuals, madmen, slaves, Jews, and women as outside the category of "human," just as it treated animal life as less

important. Humanism has been linked to Europe's imperial destiny of exclusion, exploitation, and conquest.

Feminist critics argue that gender identities are not natural. Categories such as "masculine" and "feminine" are social constructions. Viewing one's body as male or female is shaped by discourses that construct these identities, rather than natural categories. Relations between genders are based on power distribution in the social order. Feminists argue that humanism assumes men are the origin and source of meaning, action, and history (Catherine Belsey). Feminist critics see identity as multiple, fractured, relational, and enmeshed in power structures.

Humanism rejects the modern divide between non-human and human, nature and culture, instead seeing humans and non-humans as living interdependent lives. It rejects the dominance of techno-science rooted in traditional humanism. Feminist perspectives on posthumanism emphasize lived realities, multiplicity, rhizomatic (non-linear) relations, nature-culture connections, and differential modes of belonging.

Critical Race Studies and Postcolonialism

Critical race studies note that in the colonial age, the black man was not a citizen, nor even human. Black women were often seen solely in terms of their bodies, labor, and reproduction. Biological theories of race allowed the treatment of black people as animals or "abhuman," with colonized people presented in a state of frozen stasis.

Humanism has been linked to racism and capitalist modernity, particularly during the era of slavery, where people of certain races or ethnic groups were denied citizenship, rights, and dignity. Enlightenment humanism, racism, and capitalist exploitation became intertwined during this time.

Context 3: Environment and the Anthropocene

Posthumanism arises during a time of clear environmental crisis: global warming, species extinction, pollution, desertification, ozone depletion, and more. This crisis draws attention to the human causes of the sixth mass extinction and the exploitation of animals through their instrumental use.

Paul Crutzen coined the term "Anthropocene" to suggest that the 12-million-year Holocene epoch has been followed by an accelerated period of human impact on the planet, to the planet's detriment. The Anthropocene concept aligns with posthumanist thought, as human impact on the planet has produced unequal life chances for many species.

For posthumanists, the Anthropocene intersects with human dealings with the earth through agriculture, fossil fuel use, and technologies like nuclear power. Political, economic, and social systems exploit land, non-human life, and differentially organize humans (e.g., plantations, slavery, capitalism, colonialism).

In the Anthropocene, life is divided into those that matter (bios) and those that do not (zoe). Our sense of the planet is mediated by technologies, from Google Earth to geological surveys. Posthumanists examine how we view the planet through technology and literature, focusing on humanity's elevated status at the top of the natural order.

Posthumanist approaches to ecological issues recognize that we live in a post-natural world, created by biotech engineering. Hybrid beings, both human and machine, once speculative fiction, are now a reality. Popular culture addresses fears of losing human control over non-human and non-living entities, such as in eco-horror films like *The Meg* or *Lake Placid*.

There is also a focus on "deep time," a geological time scale that existed before humans appeared on Earth, sparking interest in ancestry and the evolution of life forms, as well as imagining a world without humanity.

Transhumanism

Transhumanism, a movement within posthumanism, emphasizes human enhancement. Transhumanists believe in the perfectibility of the human. They view the human body as inherently limiting and weak, and therefore believe it must be transcended through technology. Disease-free, long-living human bodies with enhanced abilities are the goal.

Philosophically, transhumanism draws on Enlightenment ideals of human rationality, agency, and perfectibility. Transhumanists seek to force human evolution, not through natural selection, but through technological intervention, to create an advanced human.

Posthumanism refers to the stage that may evolve after the current transhuman era. Transhumanists hope to eradicate biological flaws and weaknesses and enhance cognitive, physiological, intellectual, and moral qualities. However, transhumanism still adheres to Enlightenment ideals, emphasizing the human-animal divide.

Transhumanism often focuses on mind-body dualism, emphasizing the preservation of memories and personality traits over the continuity of the physical body. In this view, the body is subordinated to the mind, which is seen as the seat of personality. Cryo-preservation of the body for later revival and downloading/uploading memories into computers exemplify this approach.

Transhumanism reinforces the "machinization" of humans and the humanization of machines, pursuing human enhancement as an ethical imperative. Critics argue that transhumanism places little emphasis on the entangled, mutually dependent lives of humans and non-humans.

Popular Posthumanism

Films and literature like *Ex Machina*, *Blade Runner*, *Minority Report*, *The Terminator*, *The Matrix*, and *Never Let Me Go* explore posthumanism. These representations often feature organic/inorganic hybrids, digitized personas, trans-species bodies, artificial beings, and enhanced humans. Popular posthumanism tends to focus on future beings whose abilities surpass those of humans, often evoking fear and anxiety.

"Speculative posthumanism" is a future concept in which enhanced beings will surpass humanity. While many of these portrayals raise fears of losing human control over non-human entities, others examine the blurred lines between human and non-human, living and non-living.

WEEK 2

Critical Posthumanism. Genealogies - 1: New Materialisms

For **first humanism**, the central concern has been the binary or dualist mode of thought that creates **hierarchies** such as man/woman, non-human/human, living/non-living, mind/matter, nature/culture, civilized/wilderness, and human/technology.

In such binaries, the first term (**human, living, man**) acquires greater weight. Thus, the first term, and therefore the being or identity, becomes the dominant one in the social order, language, and modes of thinking.

For **posthumanists**, the over-importance given to mind over matter, language over material, has been problematic. According to critics like **Karen Barad**, **Vicki Kirby**, and others, everything, including matter, is turned into a matter of language. The call here is the **technologization of matter**. The argument is that **matter and meaning** are entangled.

New materialism adapts the work of a strand of feminism known as **material feminism**. Matter and materiality cannot be ignored in favor of discourse, as was often the case in **poststructuralism**. As the new materialists put it, "**matter matters**."

To **think with matter** is to have an ethical commitment to matter, whether human or non-human, whether living or non-living.

New materialism rejects the **nature/culture binary**, instead proposing a **nature-culture continuum**.

This means that **matter in nature** cannot be separated from its economic value or the symbolic meanings assigned to it by humans. New materialist philosophy covers **organic non-human life (Zoe)**, the **geological foundations of living matter (Geo)**, and **technological mediation**, producing notions such as **ZoeGeoTechno**.

New materialists argue that **matter has agency**—capabilities that produce change.

Matter and particles, whether living or non-living, acquire agency through their **interactions**. Non-living matter also possesses agency and capabilities because of its relationship with other matter.

For new materialists, there is no **inherent agency, autonomy, or identity**. All of these are consequences of relations between forms and matter. In other words, **identity and agency are relational**.

To accept **relationality** means accepting certain consequences. **Human processes, identity, and agency** are shaped by the world, including the non-human and non-living.

The human is entangled, even in terms of its very being, with other life forms and non-living matters, including technology.

All matter is therefore **collaborative**. All identity is **cross-species** and across material forms, including the non-living.

From the above, three key propositions emerge:

1. Humans cannot think of care, recognition, and compassion as directed solely at other humans because human life and material are entangled with other forms of matter, life, and non-living things.
2. **Recognition and care** must extend to these other forms of matter as well.
3. This proposition is central to **posthumanism**. All life forms and non-living matter, as well as human identity, subjectivity, and agency, depend on these entanglements. Questions of recognition and identity must include human relations with other life forms and the non-living.

We live our everyday lives surrounded by and immersed in matter. We are composed of matter ourselves, experiencing its restlessness and **intra-actions** even as we reconfigure and consume it. At every turn, we encounter physical objects fashioned by human design or natural forces that structure our daily routines for survival. Our existence depends, from moment to moment, on myriad microorganisms, diverse species, barely understood bodily and cellular reactions, cosmic motions, and material artifacts.

New materialism proposes that **being** is in fact **becoming with**. Who we are as individuals depends on our interactions with other life forms, drawing upon them, influencing them, and being influenced by them—from microbes to technology.

Power, care, and compassion flow across and through relations between matter, whether living or non-living, human or non-human, all of which are part of **assemblages**.

Posthuman ethics embraces **anthropomorphic specificity** to distance itself from the arrogance of **anthropocentrism** and **human exceptionalism**. Acknowledging the **agency of life** inoculates against the intoxicated metaphysics of life itself. Paradoxically, it is by embracing the specificity of our **anthropomorphic frame**, with its limits and possibilities, that humans can connect to non-human others. By being **grounded**, we can flow more easily toward others.

In this **relational effort**, the shared imaginary and experience of the **less-than-human** and the **humanized others** who were not allowed to be fully human to begin with become sources of inspiration.

Critical Posthumanism. Genealogy 2: Human-Animal Studies

Critical posthumanism begins by questioning the **human-animal divide**.

In terms of etymology, **animal** comes from **anima**, which means **soul** or the animating principle of life. Critical posthumanists question how, when all creatures have this anima, humans have positioned themselves as distinct and superior to animals.

Traditional humanism sees the **non-human** as the **naturalized other** of the human. It is humans who define what counts as human qualities and animal qualities. This hierarchy has resulted in the exploitation and extermination of entire species, a phenomenon known as **speciesism**.

Speciesism enables the construction of the human as the sole **rational creature**, transforming animals into mere objects, commodified as exotic creatures or meat. Animals are treated as noble but subject to the human gaze, seen through different lenses—sentimental or utilitarian.

The term **animal** erases distinctions between species, speaking of **the animal** as a single entity without the capacity to respond in **human language**. Speciesism also excludes not just non-humans but also **non-white, non-masculine, non-normative** groups (e.g., disabled, malformed, or enhanced individuals).

Critical posthumanism questions the construction and categorization of **the animal** in various domains:

- The ability of animals to experience **pain and suffering**.
- **Animal consciousness**, their sense of mortality, and their ethical capacities.
- The ways animals are represented in **literature, popular culture, and science**.

When **ethologists** or scientists reduce animals to **stimulus-response** mechanisms, humans strip animals of their agency and diminish the complexity of their minds.

Critical post-humanism Generality 2 human-animal studies 2

OTHERING animals - Common modes

Pet, monsters, spectacles, test subjects.

Animals as pets

Seeing animals as pseudo-humans, anthropomorphizing them as making up for some lack in human life, they are like us yet not like us, existing in human spaces and yet inside.

I view animals that are excluded from the home; then, the boundaries that the home symbolizes between inside and outside, friend and stranger, private and public, and so on seem

to always formally be in places where securities would appear to be assured. The existence of a group of animals that live inside the human home might be read as a challenge to these boundaries.

Animals as monsters

Predators: Speciesism is unable to encompass the horror of specific species of non-humans as monsters.

Eco-horror

The natural world, when misinterpreted or interfaced with, whether intentionally or unintentionally, is beyond human control. In cases like *The Shallows* or *Lake Placid*, humanity's interface with the ocean or lake mutates into a confrontation with the habitual residents of the terrain. In most cases, humanity leaves its cultural and geographical boundaries and enters the natural world for recreation, escape, or exploration. It reinstates the dichotomy of nature versus culture. Humanity, representing culture, crosses boundaries when it enters the world of the bear, the shark, or the crocodile.

Animals as spectacles

Modernity is characterized by the near disappearance of wildlife from human lives and their food from its vision and material experience. This means there is a marked shift in our experiences of animals. While once they were experienced firsthand, face-to-face, they can now only be experienced in the virtual spaces of representation—whether in print, photographs, on-screen, or in enclosed specific spaces like zoos and sanctuaries. In short, these bring nature, the wild, in.

Critical post-humanism argues that

- The agency of non-humans is a given.
- The animal has some moral status akin to a human.
- The vital force cannot be restricted to humans and some forms of the non-human, as it cuts across and reconnects previously aggregated species categories and domains. The human can no longer be separated from the material, including organic plants and animals, and inorganic technology and information at work. The human's very consciousness depends upon these networks. The human shares not only origins and evolutionary stages with other life forms but also mortality and vulnerability with them. Animals are constitutive of and not prosthetic to human words, human identities, and human behavior and spaces such as the home and family.

Critical post-humanism genealogy 3 consciousness and cognition studies

The first major revolution in thinking about information flows, feedback loops, and cognition came in the form of cybernetics.

Ideas about the feedback loop and new theories of cognition and information processing emerged with the work of **Heinz von Foerster** and **Ikhlas Luhmann**.

Together, these ideas not only enabled the building of larger and more powerful computers, they also informed ways of thinking about consciousness.

Cybernetic theory defines the human as an information processing entity that is given to communicate. The focus was therefore on the flows of information and its patterns.

It declared consciousness, cognition, and perception as constructed and as the effects of information flows from the environment, contingent upon the feedback mechanisms upon a process rather than material or structural elements.

The focus on information flows across human biology and the environment—machine and man—struck at the idea of a unified and self-contained human, proving that the human boundaries with the world are not sacrosanct, treating the human as embedded in a continuous feedback loop in which information flows in and out of, and across, human and environment.

Autopoiesis

An autopoietic living system has identifiable boundaries that separate it from its domain or environment, and every system strives to maintain its organization.

These properties are not the effect of the individual components of this system but rather the effect of the interactions of the components; the boundary itself is the effect of component interactions.

These interactions often change the components of the system in what is known as emergent behavior.

An autopoietic system is organized, defined as a unity, as a network of processes of production, transformation, and destruction of components that produce the components that—

- Through their interactions and transformations—continuously regenerate and realize the networks of processes and relations that produce them, and
- Constitute the machine as a concrete unity in the space in which the components exist by specifying the topological domain of its realization as such a network.

Contemporary thinking and research have moved beyond the brain-mind model of human consciousness to propose an embedded consciousness that is dispersed. For the humanist, this has been a major shift in thinking because it does away with the mind-body dualism while also refusing the decentralized mechanism of consciousness that has been the foundation of liberal humanism.

Recent research demonstrates that intelligence is not only embedded; it is also environment-dependent. Neural mini-brains work best when they are able to collate information from other mini-brains to which they are connected. Consciousness depends on the body's interaction with the world.

Consciousness is an emergent condition contingent upon the interactions of the various distributed organs with the environment, the interactions of these organs among themselves, and with the brain.

The post-humanist vision of human embedded intelligence draws its selfhood from the sum total of the interactions of its parts within an environment. Rather than self-contained consciousness, post-humanists argue that consciousness emerges within an environment and through distributed networks beyond the brain.

Consciousness is therefore the interaction of multiple components of the human and the world, even as the world is produced for the human due to consciousness.

However, research now establishes the presence of non-conscious cognitive processes operating at the level of neuronal processing, inaccessible to conscious introspection but nevertheless essential for consciousness to function.

Non-conscious cognition is the unthought.

When we learn by imitation and understand others' actions, it is possibly because of the work of the mirror neuron system. By observing another individual, mirror neurons fire, enabling one to plan or model one's action to attend the same. Often referred to as resonance, when one person's action triggers similar effects in the observer, mirror neuron activity is now being seen as a causal factor in empathy. Empathetic concerns, where one's feelings are affected by seeing the feelings of others, are perhaps rooted in such mirror neuron activity.

People around us are no longer just part of the world out there. Through shared circuits, the people around us, their directions and emotions, permeate many areas of our brains, which were formerly the safe harbors of our identity, our motor systems, and our feelings. The border between individuals becomes permeable; invisible strings of shared kinship tie our minds together, creating the fabric of an organic system that goes beyond the individual.

Should we consider then consciousness as extended, as the post-humanists do?

The emphasis on non-conscious cognition participates in the central thrust of decentering the human, both because it recognizes other agents in addition to consciousness and cognitive processes, and because it provides a bridge between human, animal, and technical cognitions, locating them on a continuum rather than understanding them as qualitatively different capacities. Non-conscious cognition concretely recognizes distinctions between different kinds of material processes and corresponding different kinds of agencies.

Consciousness is not restricted to the brain; it is the function of an organism, not an organ. The human body is not separated from its environment; neither is consciousness from the mind.

Consciousness emerges from specific conditions, the interaction between and across those conditions. Consciousness, body, and environment are all continuous.

An enactive theory of cognition proposes that cognition is not embedded in the organism alone but is the result of the organism being in an environment. Thinking is adaptive; thinking occurs because of adaptive interactions between the organism and its environment.

Critical post-humanism genealogy 4 feminist epistemology, ethics, and post-humanism

Feminist approaches to post-humanism, which we can call **feminist post-humanities**, call for an understanding of the relationship between political animals—humans and other-than- or more-than-human kinds, bodies, technology, and environment.

Feminist critics reject the Eurocentric and normative humanist ideas of man, and feminist post-humanities begin from this point.

Feminist post-humanities dethrone the man at the center. Human exceptionalism, in other words, needs to end.

Hence, their endorsement of plant studies, disability studies, and animal studies as a part of humanity.

Feminist post-humanities and difference

Feminist philosophies of sexual difference by **Luce Irigaray** and others, through their critiques of dominant masculinity, stress the Eurocentric biases of both humanism and universalism.

Difference is a keyword in feminist epistemology because, while some differences are playful, others are about power and domination—for example, racial and gender differences that cause oppression and domination of one section of the people.

Feminist post-humanities and techno-science

Feminist critiques techno-science as reinforcing historical gender inequalities and power relations, encoding an instrumental view of women, animals, non-humans, and the planet itself.

Feminist critics opt for relationality, sharing, mutuality, and co-becoming over the hierarchical and unequal gender structures of techno-science and capitalism.

Feminist philosophies of technology reject the view that devices designed, constructed, and used by humans are neutral and have no influence on human behavior.

Critical post-humanism

Critical post-humanism draws on post-structuralist theories of the human to challenge the liberal humanism notions of individualism and autonomy. It seeks to show how the human cannot be seen as self-contained or isolated.

Its advocates emphasize interconnection, entanglement, and relationality as a central concern.

Thus, its objective is to show how power flows across networks.

In order to do this, it often returns to the human to show how power is materially distributed and how it intersects with other life forms, technologies, and environments.

Critical post-humanism centers around issues of ethics, politics, agency, power, and difference.

Critical post-humanism contends that a new approach to ethics is called for, one that recognizes the agency of non-humans and develops a sense of responsibility towards them, with a view to addressing current issues, such as climate change, ecological crises, and the existential threats that emerge from them.

In so doing, the critical post-humanists draw on non-human and post-human life to challenge notions of humanity as distinct from the world and show how both human and non-human life are vulnerable and contingent.

This body of work engages with the question of whether other species should be given moral consideration and how human-animal relations can shape our understanding of human nature and our moral responsibilities.

Such inquiries lead to a more profound ethical relationship that can emerge in human-animal relations, characterized by new avenues for respect and connection with other life forms.

Conclusion

The genealogy of critical post-humanism reveals a profound ethical commitment to non-human life and a rethinking of the human as a contingent life form within an emergent world that extends beyond traditional boundaries and definitions.

Critical Posthumanism is an interdisciplinary framework that examines the implications of **posthumanism** through various lenses, including **philosophy**, **cultural studies**, and **disability studies**. It challenges traditional **humanist** ideals that center around the notion of a stable, autonomous human subject. Here are some key aspects:

Key Concepts of Critical Posthumanism

1. **Revising Humanism:** Critical posthumanism critiques the **anthropocentrism** of traditional humanism, proposing that human beings are not the sole bearers of meaning and value. It emphasizes interconnectedness with non-human entities and the environment.
2. **Embodiment:** This approach highlights the importance of the **body** and the experiences of **disability**, **gender**, **race**, and other identity markers. It interrogates how these intersect with notions of what it means to be "human."
3. **Technological Mediation:** Critical posthumanism considers how **technology** shapes human experiences, identities, and relationships. It questions the boundary between the human and the non-human, particularly in the context of **biotechnology**, **AI**, and **cybernetics**.
4. **Disability Studies Intersection:** Within critical posthumanism, disability studies play a crucial role. It challenges **ableist** assumptions and reconsiders the concept of "normalcy." Disability is not viewed merely as a deficit but as a unique way of being that can offer insights into human experiences.
5. **Ethics and Politics:** Critical posthumanism engages with ethical questions surrounding technology, governance, and social justice, particularly concerning marginalized groups. It seeks to create a more inclusive understanding of agency and subjectivity.

Genealogies in Disability Studies

Genealogy in disability studies refers to the historical and cultural analysis of how disability has been constructed and understood over time. This concept often draws from the work of philosophers like **Michel Foucault**, who emphasized the relationship between power, knowledge, and social practices. Here are some points regarding genealogies in this context:

- **Historical Context:** Genealogies examine the historical development of attitudes toward disability, the institutionalization of individuals with disabilities, and the societal norms that shape perceptions and treatments.
- **Cultural Narratives:** This approach analyzes how cultural narratives around disability influence social policies, practices, and identities, highlighting how these narratives can reinforce or challenge ableist ideologies.
- **Power Relations:** Genealogies investigate how power dynamics affect the lives of individuals with disabilities, including the ways in which they have been marginalized or empowered through various social and political movements.

What is Genealogy?

Genealogy is a method of historical analysis that seeks to understand the development and evolution of concepts, practices, and institutions over time. It involves tracing the origins and transformations of ideas and examining how power relations shape these changes. Originating from the work of **Friedrich Nietzsche** and further developed by **Michel Foucault**, genealogy emphasizes the contingent and constructed nature of knowledge and norms, rather than presenting them as fixed or inevitable.

WEEK 3

Lecture 1 - Posthuman Bodies I

The terminators and robots of films initiated the popular mythology of the cyborg (cybernetic organism) in the 1980s. The merging of the human body with machines to create a hybrid is the cornerstone of popular posthuman bodies.

However, the origins of this concept can be traced back to the medieval ages, with images of automaton-like machine men and the emphasis on the mind and body as machines, themes that emerged during the European Enlightenment.

Representations of the cyborg, in terms of biohybrid and species-crossing humans, are present across cultures: angels, zombies, shapeshifters, chimeras, etc.

This biohybrid is a border creature between human and machine—neither fully human nor machine, but both. It breaks down categorical and even species boundaries.

There is a disavowal of static adaptation, as the human material body adapts to and integrates with machines.

Chimeras (animal hybrids) are also posthuman in the sense that they transcend species borders.

The cyborg is an alternative way of thinking through the human and non-human subject.

Posthumanism focuses on the transformation of the body, based on the assumption that the age of the integrated, bounded body and identity is over. All bodies are multiple, fluid networks capable of morphing into or connecting with other bodies, including animal and machine bodies.

We live in an age of extensive cyber-organization:

- Our bodies are closely intertwined with technology, from medical and prosthetic devices to the digital world.
- With transplantation, genetic engineering, and other biotechnologies, the borders between species have blurred.
- Our bodies and identities are rendered into data stored elsewhere (also known as the "mathematization of the body").

The Body as Code

The human body has come to be seen as an articulation of a code—DNA. The chemical codes within molecular biology, the number of chromosomes in genetics, and the mapping of the entire genome sequence reduce life to a molecule.

The biological body becomes **biomedia**, which has two aspects:

1. The body as data, stored in computers and databases—dematerialization.
2. The data that can generate a body—rematerialization.

This means biology becomes a set of numbers and datasets, usable in the assembly of an identity for any purpose.

All bodies are trackable as data, from Aadhaar cards to biometric identification at airports, security checks, and passport offices.

Enhanced Bodies

Cyborg soldiers are the stuff of popular posthumanism. The human-machine-weapon system cyborg is now commonplace.

Enhanced bodies in sports are controversial, especially after the "Blade Runner" case and others.

Cyber-modification applies to non-human bodies as well, in terms of cloning and farming. Large mammals are genetically engineered to develop products such as skin, organs, bone marrow, and cartilage for human use.

New technologies, for example, allow keeping a dead mother alive as a "cyber-womb" until the baby is born.

All of this raises major ethical questions about power, bodies, gender, and ableism.

The Networked Body

In posthumanism, consciousness, agency, and subjectivity are emergent conditions.

The autonomy of the human, which supposedly marks its subjectivity, is constructed through relations and dynamics that cut across organic and non-organic actors—machines and humans.

Today, the body exists in conjunction with technology, within networked environments such as GPS, through which we connect to cities, infrastructure, access points, public spaces, etc.

The body is now an interface, a node through which information flows, and these flows are instantiated through and as the body.

Transplantation, Clones, and Posthumans

Organ transplants alter our sense of self in terms of embodied identity, self-reflection about the state of the body, the new organs inside, and social identity.

The borders and integrity of a body that has received organs from elsewhere are blurred and uncertain.

Clones disturb our sense of species identity. While derived from humans, they replicate humans, yet we struggle to perceive them as fully human. Whether such beings, derived from our DNA, are to be treated as persons in the full sense is a major debate in medical ethics, philosophy, and politics today.

Clones, as versions of humans, are akin to us but not us. They challenge the idea that they are created by us but not seen as human progeny (a theme initiated by Mary Shelley's *Frankenstein*).

The cyborg forces us to ask: What does it mean to be human when the body is composed of organic and inorganic parts, either within it as implants or linked to it externally?

LECTURE 2: Posthuman Bodies: Cyborgs 2 - Gender and Cyborgs

The cyborg, a hybrid organism, is a mix of nature and culture, being part human and part machine. It is seen as transcending traditional conceptualizations of essentialized gender

differences, gender roles, and gender identities. Since the cyborg blurs boundaries, it also disrupts fixed identities, and therefore, traditional categories such as mind/matter, body/spirit, and male/female become blurred as well.

Donna Haraway suggested that the advent of cybernetics might aid in constructing a world capable of challenging gender disparities.

The traditional association of men with the mind and women with matter becomes redundant in the cyborg.

The dualisms of self/other, mind/body, culture/nature, female/civilized, primitive/reality, appearance/part, agent/resource, maker/made, active/passive, right/wrong, truth/illusion, total/partial, and God/man are challenged in intriguing ways by high-tech culture. It is no longer clear who makes and who is made in the relationship between humans and machines, or what constitutes the mind and body in a machine, which dissolves into coding practices.

The cyborg can help us take more seriously the fluid, partial aspects of sex and sexual embodiment.

This involves the skillful task of reconstructing the boundaries of daily life through partial connections with others, and in communication with all parts of our beings.

The cyborg, as a "dangerous possibility," conjures a vision of connected humans, machines, non-human animals, and matter, offering an alternative model of the subject as interconnected, dependent, fluid, and partial.

In posthumanism and feminist responses to the cyborg, the human body is marked by openness, ambiguity, and connections—to other bodies, including non-humans, technology, and the world.

Bodies are seen as transcorporeal, linked to other bodies (Alaimo).

However, the significance attached to the cyborg as a hybrid in posthuman thought has been criticized for overlooking questions of race and ethnicity.

Anne Balsamo questions: When the human body is fractured into organs, fluids, and genetic codes, what happens to gender identity? When the body is broken into functional parts and molecular codes, where is gender located?

Feminists interpret reproductive technologies as enabling new conceptualizations of gender roles, such as motherhood. They also redefine what we mean by kinship and family, as the processes of reproduction are assigned to, and shared with, machines.

However, feminists are aware of the link between such technologies and racialized capitalism, which creates exploitative conditions for women of color through practices like surrogacy factories and infertility tourism.

While reproductive technologies have the potential to change our understanding of reproduction, family, and gender, this potential is not inherent in the technologies themselves. It must be developed, supported, and continually renegotiated through cultural, technological, and gender frameworks that challenge both material realities and discursive practices of domination, allowing for new realities of relatedness.

LECTURE 3

Posthuman Biology

Cybernetic theories, along with the work of Catherine Hills and others, treat the body as interfaced with machines. Human bodies are viewed as part of a distributed system of networks.

Posthuman biology presents bodies as networks connected with machines and other life forms. While we are autonomous beings, as the body tries to maintain equilibrium and exercise control over its cells, we are also organisms interacting with our environment. Living beings are autonomous in their biological, physiological, and cognitive processes, with the capacity to regulate themselves for survival.

The nervous system functions as an autonomous dynamic system, absorbing information from the environment and processing it through its internal feedback loops within neural networks. It responds to external stimuli, referred to as "perturbations" in cognitive theory, while continuing its internal activities.

Cognition is embodied—it does not occur in isolation but within a context that includes the biological body (embodiment) situated in an environment. Cognition represents the adaptive interaction between the biology of an organism and its environment.

Human subjectivity is techno-subjectivity because the body is embedded in and interfaces with machines and systems, including data and electronic systems. Humans are part of a larger system, sharing borders and functions while drawing stimuli, nourishment, and data from it.

Multiple forms of life constitute the human, and evolutionary traits of the future may already exist today. Human biology has co-evolved with other life forms and will likely continue to do so in the future. This implies a form of species cosmopolitanism where evolution blurs species identities and boundaries.

The posthuman perspective prioritizes information patterns over material instantiation, viewing biological embodiment as an accident of history rather than a necessity of life. It critiques the traditional Western notion of consciousness as the seat of human identity, suggesting that consciousness is an epiphenomenon or an evolutionary abstraction, not the "main event" it is often portrayed as.

Posthumanism views the body as a set of processes we learn to manipulate, where extending or replacing the body becomes a continuation of processes that began before birth.

Most importantly, posthuman perspectives configure the human being to be seamlessly articulated with intelligent machines. From this view, there are no essential differences between bodily existence and computer simulation, cybernetic mechanisms and biological organisms, or robotic technology and human goals.

Humans share ancestral kinship with various life forms. Human DNA, a product of evolution and change, is not simply a set of data—it requires interpretation and implementation in specific settings. Our biology is a fusion of the human and nonhuman, including microbes. Even the human genome is a "hologenome," a combination of genetic information from the host and its symbiotic microorganisms.

Humans are holobionts, assemblages of species living together. What is human, if there is no human without bacteria, viruses, and other organisms? - Francesca Ferrando.

Even our emotions are embodied and embedded; we are structurally hardwired for certain affective responses. However, the triggers for these responses arise from environmental interactions, social training (such as emotional intelligence), and our biological wiring.

Our affects are the result of bio-social-technical assemblages that form the human being. **Neuroplasticity** allows the brain to initiate structural and functional changes in response to external or internal stimuli. However, plasticity can also be destructive—diseases like Alzheimer's or brain lesions can result in a completely different person, a form of destructive plasticity.

The self includes both body and brain, or mind, with an array of senses such as self-reflection and awareness of interconnectedness. These interconnections are not only internal but also interpersonal, social, and political.

LECTURE 4

Biotech, Biocapitalism, and Posthuman Bodies/Biology – I

Critical posthumanism draws attention to biological citizenship, where the material body is produced and enmeshed in technoscience and capitalist processes, exploiting biopower.

Biocapitalism and Lively Matter

Biocapitalism refers to the transformation of human bodies, tissues, and organs into resources, commodities, and disposable materials. It is the merger of late 20th-century biotech with capitalism, a key feature of contemporary biopolitics, defined as the control and management of populations and biological bodies by the state and corporations.

Human bodies are treated as open to extraction, like land, by companies. It also involves creating bodies and biological matter, such as animals, solely for purposes like meat production or servitude.

Biocapitalism represents the convergence of life sciences with capitalist systems, incorporating life sciences into market regimes. In biocapitalism, it is not just individuals, but entire species and their biological materials—whether human or nonhuman—that are subject to control and appropriation by corporations, industrial laboratories, and the state.

Biotech companies own and patent biological materials, ranging from DNA to entire bodies. Biocapitalism has also produced **biosurveillance**, where organs, tissues, offspring, thoughts, and memories are databased by states or corporations. This power over life and its constituents, from tissues to memories, is held by corporate houses working alongside state laws and bioscience.

There is now a clear link between intellectual property, capitalism, and the ownership of both new and old forms of life. The development of hybrid varieties, genetically modified organisms (GMOs), and new forms of knowledge gathering have resulted in major financial investments and stricter intellectual property laws.

Patent disputes over plants, seeds, and other biological materials, which are common in the 21st century, are indicative of this link between large corporations and their projects—some of which are executed in collaboration with the military, as seen in initiatives like the **Human Genome Project** (HGP). These projects determine the fate of species, a characteristic feature of the neoliberal economy.

Judicialization of Life

The health regimens that mandate regular medication, the monitoring of health parameters, and their integration with health services, where health records are databased, are forms of the judicialization of life. Even reproduction itself becomes corporatized and judicialized under state or corporate control.

Important Comments:

- **Ethics:** A key theme here is the **ethical questions** that arise from posthumanism. As technology blurs the line between human and machine, it's important to consider how this affects things like privacy, identity, power, and inequality.
- **Future of Humanity:** This topic asks us to rethink what it means to be human in the age of technology and genetic engineering. It's about imagining new possibilities for human evolution, but also questioning the impacts on society and culture.

LECTURE 5

Biotech, Biocapitalism, and Posthuman Bodies/Biology - II

Judicialisation of Life

The health regimen that mandates the taking of **pills** and **medications**, the monitoring of **health parameters**, and their integration with **health services**—where health records are data-based—constitutes a forced quality of procedures in contemporary culture. This is driven by, say, **intellectual insurance policies**, which are state-ordered measures and form a **judicialisation of life**.

Carried forward with memories of **eugenics** from the modern era, the reproduction of life itself becomes a corporate endeavor and a judicial matter.

Science, Capitalism, and Kinship

The **Human Genome Project** and the **Human Genome Diversity Project**, conducted by organizations like **National Geographic**, **IBM**, and others, explore the origins of human relationships and **genetic citizenship**. They contribute to a new dimension in our understanding of human **identity**, **belonging**, and **kinship**.

The National Geographic **Genographic Project**, launched in 2005, aimed to employ advanced **DNA analysis** to answer fundamental questions about human origins and how we populated the Earth. The project links **genetic ancestry** with geography.

The Genographic Project assumes relatedness through **genetic unity** in the face of a historically organized social imaginary of difference and cultural hierarchies. **Kinship** based on genetic lines may thus undermine tribal self-definition and recognition.

The project implies that an individual's **ancestry** is defined through the very small portion of **genetic material** that is inherited maternally and paternally (Catherine Nash).

In the quest for origins, the project posits that **genetic origin stories** are more accurate than native myths and stories. Because the West is associated with **science**, it frames the narrative as a journey from myth to rational thought. This lack of attention to native forms of knowledge and meaning-making, in favor of the scientific history being constructed regarding races, cultures, and species, is central to its political implications.

Kinship, belonging, and a sense of **community** for **Native Americans, Asians**, and several other populations are built not on **bloodlines**, but on **cultural memories** and shared practices within their social ontology, rather than their biological roots.

Biopolitics and Biocapitalism in Contemporary Literature

Critical posthumanism is attuned to **biological citizenship**, where the material body is produced and implicated in techno-scientific and capitalist processes of exploitation through **biopower**.

Biotech and **biocapitalism** can be studied under three heads:

1. **Biocapitalism and Lively Matter**
2. **The Judicialization of Life**
3. **Science, Capitalism, and Kinship**

Biocapitalism and Lively Matter

Biocapitalism transforms human bodies—from tissues to organs—into **resources**, **commodities**, and **disposable materials**. It represents a merger of late 20th-century biotech and capitalism and is a key aspect of contemporary **biopolitics**, defined as state or corporate control over and management of populations and biological bodies. This perspective treats human bodies as suitable for extraction, similar to land owned by companies. It also involves the creation of bodies and biological materials, such as animals, for greater yield and cloning for specific purposes.

In biocapitalism, it is not just individuals but entire **species**, their identities, and their biological materials—whether human or non-human—that are subject to control and appropriation by corporations, industrial laboratories, and the state. **Biotech companies** own and patent biological materials, from **DNA** to entire bodies.

Biocapitalism produces **biosurveillance**, where organs, tissues, thoughts, and memories are all databased by the state or corporations. This represents a power head over life and its constituents, from tissues to memories, again by corporate entities working with state laws and biosciences.

Malinda Cooper argues that what neoliberalism seeks to capitalize on is not merely the public sphere and its institutions but more pertinently the life of the nation—social and biological reproduction—as a national reserve and foundational value of the **welfare state**.

There is now a clear distinction between **intellectual property capitalism** and the ownership of new, as well as some old, forms of life. The development of hybrid varieties and **genetically modified organisms** (GMOs) has led to new realms of knowledge gathering, heavy financial investment, and tight **intellectual property laws**.

Patent disputes over plants, seeds, and other biological materials, which are common in the 21st century, serve as an index of this link.

When large corporations and their projects—some executed like the **Human Genome Project** (HGP) in collaboration with the military—enter the life sciences arena, **biocapitalism** determines the fate of species. This is a specific feature of the **neoliberal economy**.

Week 4

LECTURE 1 Companion Species & Multispecies

The Question of the Nonhuman

The comparison of species and multispecies thinking in **posthumanism** consists of questions surrounding the nonhuman and human relationships.

The Question of the Nonhuman Animal

Multiple philosophical traditions have approached the animal question differently. Whether the animal is at the centre of creation, animals are seen as superior to humans, or the denigration of animals occurs because they do not have language—all these perspectives shape the question of **animal suffering**. The inquiry delves into the animal in all human nature, where the human effort is to expel the animal. The concept of the animal as possessing its own **interiority** and **subjectivity** highlights the linkage between human operations on animals and operations on other humans, fostering a relational view of human and nonhuman existence.

Symbiosis

The idea of companion species draws on the concept of **symbiosis**. **Lichens** consist of a fungus living with an algae or cyanobacteria. The non-fungal life form undertakes photosynthesis,

providing fuel for the metabolism, while the fungus enables the lichen to survive in extreme conditions. These two different life forms survive through mutual dependency.

There is also a close relationship between fungi and plant roots. In **mycorrhiza**, the fungal body infiltrates or even enters the roots of plants. Those plants that lack **chlorophyll** are entirely supported by nutrients provided by the fungi in their roots. Several fungi bore into the earth and provide the minerals necessary for plants to survive.

For **Donna Haraway**, the history of dogs, a key animal in her work, can only be told as intertwined with the history of humans. However, each species emerges from and flourishes in its own distinct material conditions and experiences.

Enmeshed Histories of Humans and Nonhuman Animals

Humans tend to see animals, especially pets, as extensions of themselves. Animals are also integrated into human-nonhuman commodity circuits, meaning the animal is socially constructed in terms of the power relations at play, such as working dogs, assistant dogs, and laboratory animals used in health and hygiene programs.

Contrary to many dominant analytical projections in the Western world that regard domestic canines as mere companions, dogs are not about projecting one's self or beauty; they are dogs. Being a pet owner seems to me to be a demanding job for a dog, requiring self-control and emotional and cognitive skills that match those of good working dogs.

Companion Species

The term "companion species" refers to a relationship that regards potential partnerships with all existence on Earth, irrespective of their biological and cognitive status—from insects to dogs and bacteria to pigeons.

Naturecultures

The idea of companion species troubles the distinction between **nature** and **culture** when speaking of humans and nonhumans. Each species has its own histories but is also intertwined with the histories of other life forms. Haraway uses the term **naturecultures** to describe companion species, emphasizing the co-becoming of different species in symbiotic relationships that share identities over years of evolution and adaptation, thus emphasizing their interdependence.

The histories of **technology** and **biology** cannot be separated due to the complex colonization of humans and nonhumans. Each species has a biological history (nature) but also a social history (culture) shared with humans. This interconnectedness represents co-becoming.

Lifeways of Species

Lifeways of species emerge from this shared condition of relationality.

Critical Animal Studies

Critical animal studies seek to move from principles and philosophies of equality and subjectivity to **political activism** around animals. It discusses the linkage between forms of human oppression—**racism**, **sexism**, and **ableism**—and what is called **speciesism**, which refers to human oppression over other species of life.

This perspective critiques the binary of nature versus culture, which has resulted in a hierarchy placing humans above animals. It examines the conversion of animals into objects for **capitalism** and **globalization**, tracing a link between discourses of inequality and the practice of animal killing. Critical animal studies demand an ethical commitment to animal rights and welfare.

LECTURE 2: The Question of the Nonhuman 2 Plants and Vegetal Life

In discussions of the **nonhuman**, while animals have received considerable attention for being on the periphery of human thought, plants are often at the margins. Some note that a tradition of Western thought, from Aristotle onwards, reflects on the **vegetal soul**. **Critical plant studies** is an interdisciplinary field that examines the relationship between plants and humans and the cultural meanings of plants.

Plant thinking refers to shifting agency away from humans and towards **vegetal life**, which is the backbone of all ecosystems. It acknowledges that discounting plant life is a grave ecological and philosophical error. Plants possess a distinct **subjectivity**. They are capable of accessing and influencing the world, just as they are open to being affected and influenced by it. Humans and plants occupy different **temporalities**—plants are governed by seasons, light, etc., but they also intersect with human life. We need to see the plants within us.

Key Questions

- What forms of thinking do plants (lacking a central nervous system/brain) engage in?
- How do humans think about plants?
- How is human thinking about plants influenced by our encounters with them?

Plants possess **non-conscious intentionality**, which means the very notion of **intentionality** is delinked from consciousness. This is a feature of vegetal life—actions taken to survive and thrive can be considered a form of **thinking before thinking**. For example, the turning of a plant toward the sun is perhaps the most iconic expression of its non-conscious intentionality. Roots also demonstrate this as they grow into the soil seeking **nutrients**. The plant's tendency to reproduce itself and its species also demonstrates **non-conscious intentionality**.

Vegetal Being

Vegetal being revolves around **non-identity**. A plant is inseparable from the environment in which it germinates and grows. Its style of living is devoid of a clearly delineated autonomous self. To be a plant is to grow endlessly, to move outward, unconfined by the self. It scatters **fruit, oxygen, and seeds** all around.

Critical Plant Studies turns to **indigenous non-European knowledge systems** to reframe the ways in which we look at plants. For example, the **Gulab Gular Bolo** people of the West Kimberley in Australia propose different relations between humans and the environment. They sustain a **cosmos of connections** among humans, plants, animals, and earth formations. In this community, humans play their part by living according to the **Dreaming Law**. If a tree is cut, its parts are used for ceremonial purposes, and humans are destined to care for trees, as they are for humans.

Sceptics, however, question whether plants have **intentional states**. They argue that we are limited by our **human-centric consciousness**, making it difficult to understand plants from outside our perspective. Classifying plants as individuals or as a confederation has also been rejected by many.

As part of **companion species** and **multispecies thinking**, critical plant studies call for a re-examination of the mutually constituted lives of plants and humans.

LECTURE 3: Companion Species & Multispecies

The term **companion species** was coined by **Donna Haraway** to describe the mutually dependent relationship between humans and nonhuman animals. She explores this relationship by noting how humans live with and depend on numerous **microorganisms** for survival, and in turn, these microorganisms benefit from their association with humans. Thus, life forms **coexist**, and we as humans need to reevaluate species boundaries and even our sense of self.

Our **self** is constituted by the other, the nonhuman within us.

Other terms used include:

- **interspecies**
- **cross-species sociality**
- **naturecultures**
- **plural relating**

Multispecies Ethnography

This field draws attention to how humans and other beings build worlds together. For instance, the mole, the rabbit, and the potato are not merely tools or building blocks for human projects, but are instrumental in world-making projects of their own.

Trees, for example, "know" when we are nearby. The **chemistry of their roots** and the **perfumes their leaves pump out** change when we are near. When you feel good after a walk in the woods, it may be certain species are "bribing" you.

When the roots of two **Douglas fir trees** meet underground, they fuse. Through these self-drafted knots, the two trees join their vascular systems together, creating one vast network underground, supported by countless thousands of miles of living **fungal threads**. Trees feed and heal each other, keep their young and sick alive, and pool their resources and metabolism. There are no true **individuals**, nor are there separate species—everything in the forest is the **forest**.

Vegetal Communications

The collective agency of plants responds to their environments and shapes them in ways that humans have refused to understand.

Multispecies Ethnography is a variant of traditional ethnography and cultural geography. It accounts for what is called **entangled lives** by:

- Examining the interactions and mutual influences of humans and all forms of life in a given territory.
- Studying landscapes, climate, and weather.

It rejects the older **nature-culture dichotomy**, showing how man-made culture cannot be separated from other life forms and ecosystems. **Human culture** has always evolved alongside nature, and nature has always been affected by what humans have built, used, or tested. Hence, the term **naturecultures**.

Shared Vulnerabilities

Humans and nonhumans share vulnerabilities. Their embodiment is embedded in specific

contexts where certain species—both human and nonhuman—are treated as **disposable lives**. The very fact of having a body open to injury links humans and nonhumans.

This recognition stems from **Stacy Alaimo's** concept of **transcorporeality**. Transcorporeality emphasizes that humans are materially enmeshed in the physical world. Therefore, **environmentalism** is not an external or optional pursuit but one that is always present and essential.

Humans and nonhumans are **enmeshed** in the same networks of power, operating both locally and globally. **Structural inequality**—based on race, class, geographical location, species, etc.—determines the fate and vulnerability of both humans and nonhumans. Depending on your species or geographic context, you can be rendered open to injury, sold, maimed, or killed.

Climate change, soil erosion, deforestation, and toxic dumping damage ecosystems for both humans and nonhumans in different but related ways. **Toxins** from the environment enter the bodies of both, linking them through **transcorporeality**.

Multispecies projects like **CMD experiments in bio-algorithmic politics** (2019) and **Microbiosis: Ancient Whose to Future Math** (2018) highlight communication across species and envision humans as part of a greater ecology. These projects demonstrate collaborations between humans (artists) and nonhumans (algae), showcasing interspecies cooperation.

Another kind of multispecies community can be seen in late 20th-century **coastal seagrass waves**, which absorb and flow with shells, plastic waste, fishnets, flip-flops, and even washed-up military waste, such as sea mines or canisters of toxic gases from previous world wars.

Lecture 4: Transcorporeal Trauma

Transcorporeal trauma is a theme in **multispecies studies**, focusing on the shared **toxicity** experienced by humans and nonhumans alike—both living and nonliving entities. This concept intersects with **posthuman studies, ecological matters, and the Anthropocene**, concentrating on the **socio-economic processes** and their material effects on humans, nonhumans, and ecosystems. It calls for **environmental justice** that encompasses all species and life forms.

Environmental Justice and Social Injustice

Environmental injustices are deeply entangled with social injustices. **Environmental justice** emphasizes the **material interconnections** between specific bodies and places, especially in

communities that have been **dumped upon**. These injustices are often shaped by **race, class, and gender**, revealing inequities based on **place-based inequalities**.

- For example, **high levels of PCBs (polychlorinated biphenyls)** have been found in the **breast milk of Nunavik mothers**, which is passed to their babies. Since PCBs are **lipid-soluble**, they accumulate in the fat reserves of sea mammals, concentrating as these animals use up their fat in winter. This can also affect **polar bears**, which emerge from hibernation with cubs already contaminated by chemicals.
- Studies also reveal that the **same carcinogens** causing cancer in humans can also cause cancer in **fish and mollusks**, showing similar metabolic processes.
- In places like **Northeast Italy**, where **illegal waste disposal** is common, both dogs and humans show high rates of **lymphoma** and **cancer**. Similarly, **military dogs** in Vietnam exposed to **Agent Orange** suffered high rates of **testicular cancer**, highlighting how toxic environments affect both humans and animals.

Shared Toxicity Across Species

Plastics and toxins serve as another example of shared vulnerability. While plastics are often described as **non-biodegradable**, certain microorganisms can slowly degrade them. The **UV exposure** and the actions of organisms, like fish biting plastic debris, play a role in this slow degradation. These processes, alongside **toxic waste** in rural areas, demonstrate how **animal bodies mediate** between toxic materials and the environment, leading to **multispecies toxicity**.

For example, **Alabama's raw sewage problem** disproportionately affects poor African-American communities, showing how the intersection of **historical injustices** and **environmental degradation** continues to impact marginalized populations.

Sacrifice Zones and Environmental Racism

The concept of **sacrifice zones** refers to areas contaminated due to the **mining** and processing of **uranium** for nuclear weapons, as well as **chemical pollution** in low-income and minority communities. **Environmental racism** often results in these communities being forced to live near toxic industries and military bases, making them disproportionately vulnerable to health and economic hardships.

- **Transcorporeal trauma** illustrates how **multispecies vulnerabilities** expose both human and nonhuman species to toxins, resulting in different yet similar consequences. These **sacrifice zones** now span the entire planet.
-

Lecture 5: Multispecies in Contemporary Literature

Multispecies belonging is a recurring theme in contemporary literature, as seen in the works of **Margaret Atwood**, **Richard Powers**, **Alexis Smith**, and **Amitav Ghosh**. These authors explore how humans and nonhumans share landscapes and dependencies, portraying **ecosystems** as **spaces of cohabitation**.

Examples of Multispecies Literature

- **Margaret Atwood – The Year of the Flood:** This novel examines how humans and other species coexist in a post-apocalyptic world, highlighting **shared vulnerabilities** in times of environmental disaster.
- **Richard Powers – The Overstory:** Powers' novel focuses on the lives of trees and their interdependence with humans. It portrays the forest as a **single organism** made up of multiple life forms, demonstrating how **other species call the shots** in shaping the environment on which all life depends.
- **Alexis Smith – Glaciers:** In this novel, landscapes are depicted as **spaces of cohabitation** where humans and nonhumans interact to survive.
- **Amitav Ghosh – The Hungry Tide:** Ghosh's novel illustrates how **species help each other survive**, particularly in ecosystems such as the **Sundarbans**, where humans and animals are deeply interconnected.

Shared Vulnerabilities and Multispecies Solidarity

Climate change has altered the behaviors of both humans and nonhumans, as seen in many contemporary works. These texts emphasize **multispecies solidarity**, where humans and nonhumans come together to face common threats.

- In **The Overstory**, the **Douglas fir** becomes a symbol of environmental protest, drawing people together to protect the forest. Similarly, in **Flight Behavior**, the **monarch butterflies** bring people together in solidarity to fight climate change.
- This idea of solidarity extends across species, revealing **multispecies alliances** that go beyond boundaries and highlight our **shared fate**. In both novels, specific species serve as **synecdoches** for larger environmental causes.
- **Interspecies solidarity** is further seen in **Flight Behavior**, where characters from different backgrounds, such as the **African-American researcher LaPido** and **Dellarobia Turnbow**, form alliances to protect both human and nonhuman interests.

These works illustrate how **multispecies relationships** are central to our understanding of survival and justice in a world increasingly shaped by **environmental crises**.

WEEK 5

Lecture 1 - Bioart

Bioart involves the use of living material and biological matter as both the site and substance of art. Biological or living material is employed with technology to indicate the linkage between nature, biology, and culture. It also focuses on the cultural understanding of the very idea of life, demonstrating how art can enhance such understanding, which was previously seen only as the realm of biologists.

Bioart interests include studying the technology of nature, genetic modification, environmental degradation, and the enormous impact humanity has had on the environment. It examines the exploitation and commodification of living matter, as well as the potential of using art to draw attention to climate crises and questions of identity and biology. These themes are subjects for elaborate art projects, ranging from sculptures to film installations, utilizing living tissue and lab-generated life in conjunction with technology and the environment.

Transgenic art is the subject of the last essays in the collection. Kac's use of synthetic genesis in organisms to create special characteristics, such as the famous glowing rabbit, exemplifies how traditionally imaginary creatures are now created in laboratories through genetic engineering.

Kac's GFP Bunny was one of the earliest instances of bioart in 2000. Bioart serves as a critique of how technology operates on living materials, drawing attention to the potential positives and negatives of such interventions. The creation of organisms as art also points to transhumanist possibilities and future routes of evolution. Matthew Barney's "Cremaster Cycle" documents paintings that represent such possibilities.

Stelarc's early experiments in attaching a third arm and an extra ear to his body were attempts to speak of biological evolution through technology. The Symbiotic Biological Art projects at the University of Western Australia, led by Oron Catts, include the Tissue Culture and Art project and the Autotroph project.

Biomass attempts to bring together life forms occurring in nature and life forms created by humans in laboratories. In stage 2, Catts and his team played epidermal and connective tissue over biopolymers and glass to show the merger of biological and lab-produced matter.

In "Crude Matter," Catts and others placed cells grown over different substances to show how various materials can produce different forms and functions of biological matter.

Bioart also often conveys environmental messages. Tissue Culture and Art's "Victimless Leather" addresses the killing of animals and their transformation into commodities, processed in factories and sold as consumer products. "Victimless Leather" was an attempt to create lab-grown leather.

Much artwork also brings together posthuman concerns, reflecting on the human impact on all life forms and the non-living. Artworks like Liliana Porter's "Water Wash," for instance, draw attention to the crisis of our waterways and consider alternative pathways for toxic dumping, among others.

Anthropocene art seeks to alter our modes of perception, which have already been modified for centuries through our lifestyles, particularly regarding climate change and environmental disasters.

Becoming a Sentinel Species

In the film, fictional researchers inject microplastics from the sea into their bloodstream and begin experiencing hallucinations and old connections to the ocean. Concurrently, real-life blood samples are taken, revealing microplastics found in everyday life. The researchers and artist observe how the toxins in the environment, upon entering the human bloodstream, can transform the human body into a sentinel, as the body starts behaving in certain ways. The hallucinations in the film and their food functions serve as indexes of the state of the environment. By employing microplastics and fictional effects on the researchers, the project draws attention to the oceanic origins of life. Here, the artist's blood is the artistic material upon which the plastics become a commentary on the impact toxic materials have on biological matter.

The project critiques species exceptionalism.

Speciesism - The hierarchy between species is created by humans, enabling them to exploit and use other species. This concomitant exceptionalism has rendered other life forms increasingly vulnerable to pollution due to human actions. The project asks why humanity should not serve as indicators of toxic biomes.

Tonn, Leslie, and Vallejo explore how microplastics enter and impact the human immune system, particularly macrophages, in real life. They also demonstrate how we are linked to other species and life forms through a shared toxicology; microplastics connect marine life forms and humans. Environmental toxification cannot be deemed a problem limited to certain species. Other-than-human beings have been drawn into the drama of environmental toxification due to human actions.

Becoming Human Sentinels shows continuous alignments of species within the larger crisis of environmental toxification.

LECTURE 2 - Posthuman Performance Art

Among the earliest practitioners of art that adopted a posthumanist and transhumanist approach was Stelarc. He added an extra arm and wrote a word with all three hands—an evolution of the concept.

Posthuman performance art focuses on the rise of composite bodies of the human and the non-human. The merging of human and non-human elements produces a third composite body with new capacities of form and function. Posthuman performances give more space and importance to things not necessarily created as objects; humans bestow value and utility upon things by making them objects.

In traditional performances, the movement belongs to humans, while objects are merely props. In posthuman performances, this prop function is modified so that agency and movement do not necessarily proceed only from humans.

Posthuman performance art calls into question the anthropocentric views of space, objects, and agency by introducing non-human elements—machines, plants, animals, and other objects—to demonstrate the connections and alignments between humans and non-human others.

Laurent Liefvooghe, Living Machine

Living Machine is both a theatre box and a fully equipped home. Here, one must perform living. Whether or not its inhabitants live there, the house continues its own life—opening the blinds, making coffee, and switching on the lights in the kitchen when morning comes. It goes completely silent and dark in the evening, etc. According to the times of day, it becomes transparent, opaque, or effective. When a person moves inside or outside the box, they are looking at others who perform. The house and its objects seem to have a life of their own.

The house from Living Machine is reused in their Dreamhouse. The real house is now set up in a museum space, so the functional house exists between a house and a museum. Everyday life becomes a performance for spectators.

Space becomes saturated with a presence that, in its absence, reveals autonomous activities and moods of being when things are left alone.

Maximillian Haas and David Weber Krebs - Balthazar

Balthazar is a fictional name for the protagonist, a donkey. The animal is placed in a group with humans who seek to engage him or her in theatrical collaboration. The donkey is at the center of the action, moving freely through the stage space while the humans are formalized into a group. At one point, they play music and listen wordlessly. They collectively decide upon the direction and draw diagonally through the entire space. The donkey follows or may just stand

still. The humans return to him, perform a few dances, stand in front of him, and then start over.

This intentionality of movement belongs solely to the humans. Balthazar is about interspecies performance, complicating the orientation of humans and non-humans in space.

UNKNOWN FIELD DIVISION - Rare Earth Metals

Rare earth metals are essential components in high-tech electronics and are used for the production of magnets, camera lenses, and batteries. They are among the world's most sought-after materials. The production of one ton of rare earth metals generates more than 8 kilograms of fluorine, 75 cubic meters of toxic wastewater, and approximately 1 ton of slightly radioactive mud.

The team members illegally entered the mining area and secretly took photographs of the mines and the tailings and labs. They extracted mud from the tailings, from which they made three vases of different sizes. Each vase represents precisely the amount of radioactive mud produced as a byproduct of the manufacturing process for a smartphone, a laptop, and the batteries used in an electric car, respectively.

Rare Earth draws attention to materiality, capitalism, global industry, and the earth. Martina Ruhsam displays the entanglement of the history of these artifacts with socioeconomic relations, the dictates of profit maximization, the international division of labor, and the conception of needs of an urban population. Moreover, it becomes apparent in the film that the history of the vases is interwoven with the specific ecological afterlife that will outlive the lifespan of the vases, as well as those who produce them. Further, the future and past are inextricably interlinked with one another.

In **Time Loop**, the collective asked participants to reflect on the deep past, a practice that the collective borrowed from the Native Indians of Canada.

Conversation Without Words is a performance that explores silence and was inspired by the annual gathering of nude chieftains who sit together in complete silence for hours to connect with one another.

The point is to question how knowledge is built, especially through the slow erasure of older forms of knowledge by Native Americans and others. It shows that knowledge is co-produced. The idea is also to explore a different kind of relationship with time, shifting focus entirely from the past and future to the present.

LECTURE 3 - Literary Posthumanism

More critics now trace the **genealogy of first humanist thought**. Examining literary texts from the **European medieval** period to the present, they find the presence of **monsters**, the **misshapen**, the **alien**, the **machine**, and other **non-normative humans**, indicating an interest in other forms of **human interaction** with the **environment** and the role of the **non-human**. Literary texts from **Beowulf** to the contemporary are therefore examined for their interrogation of the **boundaries of the human**, the **hybridization of the human**, and the **idea of nature itself**.

Milton, **rebellious** and other **European writers** rejected **anthropocentrism**, celebrating **animal-human hybrids** and the **non-human**. This early expression of **posthumanism** is also seen in the **visual arts**, notably in **Titian's works**.

Many authors in the **medieval** and **early modern European period** were also interested in the **non-normative human form**, expressed in themes of **disability** and **monstrosity** (hybrids), e.g., **Grendel** in **Beowulf**.

Beowulf and the **medieval monster**, in conjunction with contemporary ideas about **disability**, suggest that the **non-human** and the **differently embodied** might model other ways of inhabiting the world. The **hero** and the **monster** are equivalent in their more-than-human **virility**: each has the strength of **30 men**. The hero must be something more than human to combat the monster.

The **monster** serves the cultural function of allowing the **human** to carve the very notion of **civilization** from a landscape perceived as **corrupt** or **evil**, a kind of **humanist project**.

Shakespeare's humanism and **troubling of the figure of the human** were accompanied by attention to **non-human others**.

Shakespeare **blurs** the human-non-human boundary:

Caliban is a **fish-human hybrid**, a **human-animal hybrid**. **Prospero** and **Miranda** refuse to enlist him in the ranks of **humanity**. He must remain a **villain**, a **poisonous slave**, a **hag-seed of malice**, and finally a **thing**.

The **artisan performers** in **A Midsummer Night's Dream**, the **rude mechanicals**, are associated not merely with stupidity but with a lack of **reasoning ability**, linked to **animality**. **Bottom** embodies the human-animal distinction but also crosses the **human-fairy boundary**. There are also hidden hybrids, where characters are tainted with the **impurities of nature**.

Troilus and Cressida's Hector is described as not merely one brute, but many rolled into one. **Alexander** describes him as **valiant as the lion, slothful as the bear, and slow as the elephant**.

"To see a world in a grain of sand and heaven in a wildflower, hold infinity in the palm of your hand and eternity in an hour." – **William Blake**

William Blake's Auguries of Innocence captures the **entanglement of space, time, and matter** that marks the **posthumanist vision**.

Frankenstein offers a view of **male birth**, with **Victor Frankenstein** giving birth, enabling us to see **posthuman reproduction** as separable from the **maternal body**.

The **creature's body itself**, as a **posthuman body**, situates itself along the liminal zone between species and between **human and non-human communities**.

Wonderland underscores the **caterpillar's unique post-identity** and signals **Gardner's warnings** about eradicating species due to **misidentification**, admonishing readers against transforming **Wonderland** into a **homogeneous nation of sympathetic humanoids**.

Antecedents to posthuman fiction: Cyberpunk

There has been a **posthumanist vision** in several **20th-century texts**, even before the advent of the term.

The antecedents of **posthumanism** lie in the **genres of science fiction**, especially **cyberpunk**, with authors like **Philip K. Dick, William Gibson, Bruce Sterling, Neal Stephenson, Rudy Rucker, and David Brin**.

The **anxieties** over **AI** and computers taking over mankind and the world were popularized in **Gibson's Neuromancer (1984)**, which also coined the term **cyberspace**.

"**Cyberspace**: consensual hallucination experienced daily by billions of legitimate operators in every nation by children being taught mathematical concepts. A graphic representation of data abstracted from banks of every computer in the human system. Unthinkable complexity. Lines of light range in the non-space of the mind, clusters and constellations of data. Like city lights receding."

Isaac Asimov's fiction did much to popularize and question the then-new science of **robotics**, which eventually became the **mainstay of much 21st-century popular cinema** and fiction.

Cyberpunk showed the merger of **machines and humans** but also demonstrated the **hegemony of technocapitalism**, where capital makes use of technology to dominate **governments and societies** and control **individuals**.

Contemporary science fiction and speculative fiction's posthuman themes include:

- The role of **multiple** and **fluid identities** (cyborgs, hybrids)
- **Artificial beings**
- **Transhumanism**
- **Dehumanization** of the human with technology
- **Posthuman care**
- **Artificial reproductive technology** and the **reconfiguration of gender**, especially in **feminist science fiction**
- **The Anthropocene**
- **The post-natural world**

Much science fiction and speculative fiction is **dystopian** in nature.

Artificial beings

Kazuo Ishiguro's *Klara and the Sun* examines **posthuman forms of care** and raises key questions like: What should be human ethics towards **artificial beings**? Are they to be treated as **substitute humans**? Is it ethical to create artificial beings to **serve us**, thereby creating a new **slavery**? When **artificial beings** resemble humans and have the same qualities, is it ethical to deny them **personhood**?

Klara's love is generous, open, and benevolent. Ishiguro contrasts this with **mother's love**, which is anxious, selfish, and overprotective – **Lanlan Du**.

So, is **posthuman care** by a robot or **artificial being** better than care by humans?

Transhumanist and **dystopian novels** often represent a loss of **present**, whether **bodies** or **relationships**, all of which are placed under suspicion or made subordinate to **machine memory** with the hope of **revival in the future**. However, **transhumanism** is at odds with the **eco-critical vision of the Anthropocene**.

Don DeLillo's *Zero K* presents a **transhumanist vision** of a future where one can be brought back to life to enjoy a new existence.

LECTURE 4 - literary posthumanism 2 - feminist speculative fiction

Case Study - Octavia E. Butler

Themes:

- From the **monster** to the **posthuman**

- Re-educating the human
- **Biological addiction** as partnership
- Redefining the human

From the **monster** to the **posthuman**: **Nathak's eyes** were narrowed by the conversion of the **epicanthic fold**. His fur grew longer and thicker on his head and around his neck and shoulders, forming a kind of **mane**. Even his face was furred all over, though the fur was short. His face was long and flat, and his body and limbs were humanly proportioned. He was not human. The missionaries saw him and his people as strangely colored human beings.

Monster-posthuman: The **memory encounter with the monster** relies on **physical appearance** to make sense of the **other's body, sex, and appearance**. Perceived human-like characteristics make it easier to accommodate the monster because the **unknown** can be compared to something or someone known.

Posthumanism as partnership: **Partnership** is giving, taking, learning, teaching, offering the greatest possible benefit while doing the least possible harm. **Partnership** is mutualistic **symbiosis**. Any entity or process that cannot or should not be resisted or avoided must be **partnered**. Partner one another. Partner diverse communities. Partner life. Partner the world that is your home. Partner **God**. Only in **partnership** can we thrive, grow, and change. Only in **partnership** can we live.

Redefining the human: **Self** is **body**. **Body** is **perception**. **Self** is **thought, memory, belief**. **Self** creates, restores, learns, discovers, becomes. **Self** shapes and adapts. **Self** invents its own **reasons for being**. What becomes of the human when we think beyond the confines of the human body and shape ourselves accordingly?

The **posthuman** opens up new ways of thinking about what **being** means.

WEEK 6

LECTURE 1 posthuman subjects 1

Posthumanism as a school of thought calls into question the supposed autonomy of the **human** and the self-founded nature of the **human**. It sees the **human** as co-evolving with **technology**, the **non-human** being, and entangled with them. The **human being** is not the center of the universe.

Post-objectivity is a **rhizomatic network** interlinked with other **humans**, the **non-living**, and the **non-humans**.

Subjectivity itself is a collective assemblage that encompasses **human** and **non-human actors**: **technology mediation**, animals, plants, and the planet as a whole.

The subject now: In the **digital era**, **humans** are **networks**. Our **identity**, even our **cognitive processes**, are merged with the devices we use, the data we share, and the feedback we receive from the **technological network** we participate in. Our sense of **location** and **direction** is mediated. Our **memory** is increasingly assigned to devices (**prosthetic memory**). The **self** is now made up of our **bodies**, our **online presence**, our **work**, and our **socializing** that occurs online. These **digital presences**, our sense of **belonging**, and our **work** shape who we are.

Our **subjectivity** is **posthuman**, now mediated by the constant **technological roles** we play, the **actions** we engage in, and their effects on our **perceptual apparatus**.

New forms of **human** force a redefinition of **subjectivity** and the **human subject**, for example:

- What is the **subjectivity** of persons with **corrective prosthetic implants** and other **technological interventions** that help them lead normally functioning lives?
- What is the **subjectivity** of persons with **enhanced abilities** due to **prostheses** and other interventions?
- What is the **subjectivity** of persons in **persistent vegetative states**?

If the **human subject** is defined traditionally as **autonomous**, are the above persons **autonomous**?

Recognition:

- We coexist with various forms of **non-human life**, including the **bacteria** in our stomachs.
- Our **agriculture** and lives depend on various **non-human life forms**, including fungi, plants, and animals.
- The **soil**, **air**, and **water** we depend on contain **chemicals** and **non-human life forms**.

This recognition, that **human existence** is dependent upon other **life forms** and even the **non-living**, calls into question what we understand as the **human subject**.

We know that our **bodies** and **life processes** are dependent upon **substances**, whether **chemicals**, **enzymes**, **glandular secretions**, **neurotransmitters** that control our **mood** and **brain activity**, or **calcium** that makes up our bones and teeth.

Our **bodies** share more with the **oceans** and the **seas** because the **human body** is principally made of **water**.

The **human body** and its various processes depend on **inanimate matter**, which makes up our **bodily matter**.

Human subjectivity is **intersubjective**: A way of the **self** as an expansion outward towards other **objects** and **material matter**.

Transjective: A being constituted by its dynamic **subjective** and **material elements**, one that emerges on the basis of this dynamic **flux**. Here, the subject's **embodiment** is linked to other **materials**—a **material entanglement**.

This also makes the **human** ethically linked to others.

Their **cells**, **bodies**, **entanglements**, **locations**, **temporalities**, and ways of existing form a collective **subjectivity** and **materiality** that will vary to a great degree.

lecture 2

posthuman subjects ii

Versions of the **posthuman subject**:

- **Humans** with **prostheses**, **implants**, **transplants** connected to **machines**, modified and enhanced through **genetic** and other **technology**.
- **Clones**.
- **Humanoid robots** and **artificial beings**.

The **moral status** of **artificial beings**: We have two tests to determine the **moral status** of any being:

1. **Personality**: If a being matches the **human capacity** to use **language**, we assume it's entitled to a **moral status** analogous to that of a **human**.
2. **Feelings and emotions**: Here, we assign **moral status** based not on **intelligence** but on the capacity to **feel** and **suffer**. For example, while **robots** do not feel **pain** even when injured, **animals** do, so it is easier to assign **moral status** to **animals** but not to **robots**.

If the **posthuman clone** or **artificial being** demonstrates:

- **Altruism**,
- **Care and sympathy**,
- **Reasoning**,
- **Sense of humor**,
- **Sentience** in general,

These fulfill the tests of **reason** and **feeling**. Would such a **robot** or **being** be entitled to the **moral status** of a **human**?

The **moral standing** of **artificial beings**: If an **animal** that undertakes **moral obligations** toward other **animals** places a **moral obligation** upon **humans**, then any **being** which undertakes similar acts places a **moral obligation** upon **humans** to recognize that **being** as possessing **moral status**.

If the **artificial being** is produced by **humans** to mimic **human feelings**, **needs**, and **behaviors**, what stops us from recognizing them as **humans** too?

If any entity performs tasks that embody the above, such as **caring** for a **human** or **animal**, making **enthusiastic remarks** or **gestures**, making **altruistic gestures**, or **learning** from its surroundings, it is entitled to a **moral obligation** from other **humans**, and we confer a **moral status** upon that **being**.

Then why not extend this to a **clone** or **robot** who does the same things?

The **posthuman** as **ersatz moral person**: If advanced **artificial beings** could engage in activities analogous to those **humans** engage in, and if these activities were to create obligations when performed by a **human**, then they would create an indirect obligation to the **artificial beings**.

Presuming **moral statuses**: Can we presume the **moral status** of such **posthuman subjects**?

Is it morally preferable to err on the side of treating a **being** better than its **status** justifies, rather than worse? Treating a **clone** or **artificial being** as though its feelings might be hurt is better than treating it badly.

What if the **artificial being** does possess the capacity to **suffer**?

Would we subject a **human person** to undue risks knowing they can suffer? If the answer is no, would we risk exposing an **artificial being** to such a risk?

If we presume that those who hurt **animals** may be open to hurting other **humans**, then we have assigned an **ersatz** (substitute) **moral status** to the **animals**. **Ersatz moral persons** are those who give reasons for being like **humans**. If **humans** demonstrate any emotional response to **humans** who are in a **coma**, whose **emotional** and **rational behavior** is different from what is socially acceptable, and whom we believe are still entitled to **human rights**, then why not assign such a status to **moral persons**? If **humans** demonstrate any emotional response to such **ersatz moral persons**, including **animals**, then why not extend this to **artificial beings**?

Allo-human persons: When **artificial beings** perform the same function as **humans**, they are extensions of **humans**. They are **allo-human persons**.

lecture 3

Posthuman subjects and Ethics 1

Posthuman subjects and **ethics** can be best understood in two principal ways: the **ethics** of employing **robot carriers** and **artificial beings** in **elder care**, **care of children**, and other such human patients; and the **ethics** of humans towards the **robot care** and **artificial beings**.

The ethics of employing robot carriers and care

More and more **robots** are now a part of the **nursing** and **care industry**. They are being employed to take care of children with **special needs**, the **chronically ill**, the **terminally ill**, and the **elderly**.

Report:

The **KASPAR robot** is used to teach children on the **ASD spectrum** to engage in **collaborative play**. The authors report that children with **autism** were more interested in and entertained by the **robotic partner** than the **human confederate**.

This poses new ethical questions around the subject of **posthumans in care**.

Concern One:

The human patient may believe that they have a **reciprocal emotional relationship** with the **humanoid robot carer** and assume that the **humanoid robot carer** really cares for them. Likewise, for children with special needs, such as children with **autism**, who are already vulnerable to social relations and attachment issues, this deceptive suggestion of care or friendship can be potentially hazardous. According to some critics, this employment constitutes deception and a loss of dignity for the human.

But should the internal shade of affection and emotion be a prior condition to the **care relationship**?

Is the **care relationship** determined by a **care environment** that is formed by gestures, movements, and articulations that express attentiveness and responsiveness to vulnerabilities within the context?

Concern 2:

When introducing children and the elderly to **robot carers**, the ethical question is of creating conditions of **risk**.

A person accepts risk because they believe the **robot** can perform a function that it cannot, or the person accepts too much risk because the expectation is that the system will mitigate the risk.

Concern 3:

Human-robot interaction hinges on established trust in **human-human interactions**. **Human interaction** depends on assumptions of **co-dependency** and the mutual risk that the other may not fulfill their obligation, being open to deception but also bound by a moral obligation.

Studies show that strong emotional connections, not limited to trust, are felt by humans with **robots** in ways that resemble or approach those found in **interpersonal interactions**. But can we use the same notions approached in **human-human interpersonal interactions** to think about **human-robot interactions**?

If proper interpersonal trust requires **moral agency**, **intention**, and **consciousness**, then a reasonable and measurable approximation of concepts of **agency**, **consciousness**, and **intentionality** will need to be applied to **human-robot interactions**.

This means unless we have methods of ascertaining the **agency**, **intention**, and **consciousness** of the **robot**, it will be difficult to think of trust in **robot carers**.

LECTURE 4

POST HUMAN SUBJECTS AND ETHICS II

We need to examine the fundamental problems of how different **posthumans** will express what they are, what **dignity** they will be accorded, and what the meaning of **freedom** is, along with how it will be guaranteed.

In 2015, communications studies showed that scholars Frog and David Smith created a solar-powered **hitchbot**, which was an iPhone enclosed in a fancy and attractive **anthropomorphic** casing. The **hitchbot** was sent hitchhiking and made it across Canada, the Netherlands, Germany, and finally into the USA. **Hitchbot** not only met a lot of friendly humans but also went viral on social media, gaining more than 35,000 followers on Twitter, 12,000 on Instagram, and receiving 48,000 likes on Facebook—all in less than four weeks.

The purpose was to see how humans would interact with the humanoid. While it did not exhibit reasoning or feelings—the two standard test criteria that enable humans to determine if the device can be treated as a non-human person—humans who encountered it largely created it with curiosity and a degree of friendliness. Some took it to a wedding, others to a comic convention. **Hitchbot** was also taken to a baseball game and even on an ocean cruise.

For some, the treatment the **hitchbot** received was an index of human society's distrust of artificial beings and robots. **Anthropomorphic** and **endearing**, responsive, and cute as **hitchbot** was, it was a key to humans in many ways. So, what should be the human attitude towards humanoid robots? Is there an **ethics** towards **posthuman** forms?

The first question is how to classify the humanoid robot. Philosophers suggest that humans or robots can be classified as **ersatz** model persons and **allo** humans. When artificial beings perform the same functions as humans, they are extensions of humans. **Ersatz** moral persons are those who give the illusion of being like humans.

The question of rights, ethics, and justice of humanoid robots can only be addressed within the **socius** for which they are designed. The question of ethics towards humanoid robots has two parts:

- Should humanoid robots be created with the sole purpose of serving humanity?
- Does it mean we humans are creating a new class species of **slaves**?

These questions themselves raise two problems.

Problem 1: If we concede that it is ethical to create and design beings meant to serve humans, whose very capabilities are directed at serving humans, then principles of **justice** demand that they be allowed to fulfill their capabilities.

If the humanoid robot is to grow and reach the peak of its capabilities, then the human social order and legal systems must allow it the rightful **freedom** to serve humans.

Problem 2: If we believe we ought not to design such beings in the first place, then it means that humanity has created conditions for a new species of **servitude**.

Once we create such humanoid robots, we have to allow and facilitate the goals, desires, and capabilities the humanoid robots were designed with. This is a **rehumanizing** of the machine by ensuring its potential is fully realized and its purpose served.

Problem 2: If we believe we ought not to design such beings in the first place, then it means that humanity has created conditions for a new species of **servitude**.

Engineering robot **servitude** entails the building and employment of non-human persons who are designed to do tasks humans find unpleasant or inconvenient. It seems possible to design robots from scratch so that they want to serve us in particular ways. In such cases, the robots are not slaves since they are not working against their will.

Once we create such humanoid robots, we have to allow and facilitate the goals, desires, and capabilities the humanoid robots were designed with. This is a **rehumanizing** of the machine by ensuring its potential is fully realized and its purpose served.

Lecture 6

POSTHUMAN VULNERABILITY, ETHICS, AND CONTEMPORARY LITERATURE VULNERABILITY AND LITERARY CULTURAL STUDIES

Vulnerability in human and non-human life forms is communicated best through literary and cultural texts, whether in novels or disaster films. **Vulnerability** is mediated for us through specific modes. Literary and cultural texts enable us to understand vulnerability because:

- They show us individual and collective bodies being rendered open to injury.
- They offer descriptions of systematic, intrinsic, or social contexts in which these injuries occur. The developed language is a representational vocabulary through which we can speak of vulnerable peoples, groups, and life forms.
- They show us how individuals and groups respond to conditions of vulnerability and thus generate hopes or despair, depending on how they imagine these responses.
- They suggest alternatives to the current scenario where new world orders, social relations, government, and attitudes can be imagined—a new social imaginary.

Literary fiction about post-human vulnerability and care asks the following questions:

- If an artificial being undertakes and performs all the tasks of a human, does it create a moral obligation to recognize it as a **moral** person?
- Is it ethical to create vulnerable post-human beings to serve humans?
- Would the creation of such beings not be a reiteration of the **slave** system?
- Are the artificial beings persons?
- What are the rights of such persons?

Ruth says in *Never Let Me Go*, "I was pretty much ready when I became a donor. It felt right. After all, it's what we are supposed to do, isn't it?"

Kathy and Tommy in *Never Let Me Go* and Dorrit in *The Unit* are reduced to organs, and their functions are determined by the social order and authorities. Their bodies are alienated from their selves because they do not possess essential control over the futures of these bodies.

The clones are vulnerable post-humans because of the laws of control and agencies over internal organs' functions, such as reproduction and sex, thereby raising questions about what it means to be human.

So, the questions Ishiguro raises here are:

- Can we or should we create live bodies whose purpose is to serve humanity?
- Is it ethical to create a new order of being, such as clones, to serve our human needs, whether as objects for sexual gratification or as organ donors?
- Are clones persons, and if yes, do they merit human rights?

Klara's moral standing as a person is structured around the theme of **care** for Josie within the ecosystem of the family. She is a humanoid robot that resembles a human being and performs in a human-like manner in an environment designed for the capabilities of the human body.

Klara believes she should try to please the family members as a way of keeping the family dynamic smooth. She detects emotional undercurrents in familial relations and adapts to them.

Klara gives up her body, her **ki** body fluid, to destroy the machine as an act that she believes will save Josie. This is a sign of moral bio-enhancement.

Klara's robot subjectivity is constructed as intersubjectivity with humans. Ishiguro implies that all her learning, empathy, and attitude stem from her social relationships within the **socius**.

Klara exhibits:

- Empathy
- Curiosity
- Altruism
- Care
- Compassion

Do these qualities make her near-human?

The questions Ishiguro raises here are:

- Are robots property, or are they sentient beings entitled to kindness?
- Is robot service acceptable?
- Are humanoid robots replacing immigrants and migrant workers, and if so, what are the ethics of such a transformation of the social order?
- Is it better or worse to have robot caregivers rather than human caregivers for children and older people?

Klara is an **ersatz** moral person.

Since Klara has undertaken to care for humans and has performed tasks analogous to what humans do, then Klara deserves human rights and privileges as an **ersatz** moral person.

Ishiguro shows how humans fail to grant this status to Klara. When unlimited humans like Rick and artificial beings like Klara are performing acts of care for other humans, the humans have an obligation to treat **ersatz** moral beings with the same rights and respect as humans.

In the interaction, the boys proceed to misbehave with Klara. Danny grabs her tightly, and the others ask that he throw her over, at which several girls, Josie included, giggle.

Ishiguro makes two points in post-human ethics: those who mistreat animals will at some point ill-treat humans. If humans mistreat those deemed or perceived as lesser than them (e.g., Klara) and beings who are among the unlimited, they will react to Klara, who is an artificial being.

If Klara's identity rests in serving humans by design, then would it be an article of injustice to let her serve humans, worry about them, and make altruistic sacrifices for them? Would it not be a fulfillment of her very design and desire to let her serve, worry, and make sacrifices? It must be noted that Klara does not feel anything other than happiness at serving humans. If humans have created a terminable being with specific desires and tasks, then would it not be just to allow the being to achieve these set goals?

However, is it ethical to create beings with the primary purpose of serving humans? Is this service to humans equivalent to **slavery**? The humans create a new caste of beings to meet their needs—be they romantic, emotional, labor, or medical.

Would this mean **human** dominance and superiority over beings that resemble **human** life forms? Is it ethical to cultivate such **servitude** and deny moral rights?

If so, we must seek to reimagine the social order. The ethical consequence for creating humanoid robots is not about whether they should be created in the first place, but how we respond to their creation. How do we grant them moral status and offer them dignity and freedom to express their lives?

We must have **human** responsibility towards humanoid robots and ask ourselves if we can live together in a just and ethical manner, with dignity, recognizing the rights and potential of humanoid beings. We need to stop being **anthropocentric** and accept the reality of **posthuman** beings.

WEEK 7

LECTURE 1 - The Environment and Posthumanism

Posthumanism and the Environment/Anthropocene

For **posthumanists**, the **Anthropocene** is an ethical concept that proposes a new era for the Earth—an era marked by human intervention into the planet, which is the direct result of:

- **Anthropocentrism** and **humanistic supremacy**, which include the belief that the planet Earth exists for humans, and that human beings are the center of the world, including all life forms.

Posthumanism argues that the environmental crisis cannot be separated from human practices and belief systems such as **extractivism**, **colonialism**, **slavery**, **monotheism**, and **capitalism**.

Posthumanism stands on connectedness, **relationality**, and **multi-species belonging**. It decenters the human in response to the environmental crisis.

Posthumanism responds to the environmental crisis by calling for whole new ways of thinking about humans and the planet, promoting:

- An ethical response or what it calls **responsibility** to the Earth and all species and non-living forms.
- Imagining, in **literature**, new ways of being in the world by thinking about the lives of other species and their role in human life, especially in **genres** such as **ecological fiction** and **science fiction**.
- Recognition of human subjectivity as **intersubjectivity** that brings the non-human into the loop.
- Retrieval of **alternative knowledge systems**, such as **indigenous knowledge**, that help decenter **Western pedagogy**, which has centered on the human.
- Modifying laws to include the other-than and more-than-human.
- Rethinking the planet Earth in terms of **sustainability**, **biodiversity**, and **linkages**.
- Legislation on practices like waste dumping by accounting for more-than-human losses, enabling new practices of care for the planet as a whole.
- Addressing the links between **patriarchy**, **class**, **race**, and **technoscience**, all of which have contributed to the environmental and **Anthropocene crisis**.
- Acknowledging other-than-human agency.

Posthumanism, Art, and the Anthropocene

Bio art is a **posthumanist** response to the commodification of nature, the centering of the human, and the anthropogenic roots of the environmental crisis.

Posthumanism-inspired art projects draw attention to the crisis while offering new ways of thinking about the relationships between the human, non-human, living, and non-living. **Bio artists** recognize other-than-human organisms as collaborators, offering fresh perspectives that recast **methodological** and **socially engaged research practices**. Including other-than-humans in research-oriented explorations of reality might guide us toward new ways of thinking that increase **entanglement**. As the life-sustaining boundaries of planet Earth are being crossed one after another, sustainable transformation is urgently required. **Bio art** can pave new pathways for reconnecting with the natural world.

LECTURE 2 - The Posthuman, the Anthropocene, and the Post-natural in Contemporary Literature

Climate change fiction, including **eco-apocalypse** and **ecotopia**, is a key **genre** of posthumanist fiction, with a strong interest in the **Anthropocene**.

Examples of this genre include **Margaret Atwood's MaddAddam Trilogy**, **Paolo Bacigalupi's Windup Girl**, and **Becky Chambers' A Psalm for the Wild-Built**.

Posthumanist Anthropocene fiction refuses to place the human at the center of the action but shows the intersections of human, non-human, living, and non-living **ontology**. It embraces non-human perspectives and seeks to communicate the suffering and subjectivity of the non-human as much as can be imagined. An example is **Jeff VanderMeer's The Strange Bird**, where the sensations and suffering of a biotechnologically modified bird are the focal point.

This fiction also acknowledges the **local** as the immediate site of **climatic crisis**, while demonstrating that the crisis may originate elsewhere. Events occur on a global scale rather than just a local one.

For example, **David Mitchell's Cloud Atlas** shows how events in one part of the world impact entirely different regions. Similarly, **Amitav Ghosh's** recent works, both fiction and non-fiction, like the **IBIS Trilogy** and **Gun Island**, show how colonizing and globalizing processes affect land, resources, and people across the planet. New slavery novels such as **Isabel Allende's Island Beneath the Sea**, **Yaa Gyasi's Homegoing**, and **Cynthia McLeod's The Cost of Sugar** show how European cultures of travel and sugar consumption drove plantation cultures in the **New World**.

Posthumanist Anthropocene fiction also addresses "**deep time**"—a time before humanity, or at least its antiquity, and sometimes centuries into the future. In many cases, there is a forced return to a more basic life, back to the Earth. Novels like **Octavia Butler's Xenogenesis Trilogy** and **Atwood's MaddAddam Trilogy** explore this theme.

These works also reimagine human spaces, focusing on elements like **water** and **air**, emphasizing commonalities and **relationalities** between humans and the environment. For example, in **Larissa Lai's Salt Fish Girl**, the creation story of the deity **Nova** centralizes the role of water in the origin stories of life forms and nations, while also questioning myths of single origins and authenticity.

Posthumanist Anthropocene fiction also explores human-made ecological crises and human attempts to overcome them. In **Don DeLillo's Zero K**, natural disasters, overuse of technology, and terrorism threaten the planet, and a cryonics facility offers a **techno-utopian** escape plan by freezing bodies to be awakened in the future.

This genre also delves into themes like **species death** and the **post-natural**, where even wilderness is constructed by humans through technology. In **Jeff VanderMeer's Southern Reach Trilogy**, the wilderness, now inhospitable to humans, is a product of **global capitalism** and **technology**.

The creatures in **Borne** are lab-manufactured, while elements like river water and soil have been irreversibly toxified. Similarly, in **Atwood's Oryx and Crake**, both humans and non-humans, including plants and animals, are technologically modified.

Posthumanist Anthropocene fiction focuses on

- planetary **precarity**,
- the **relational** and connected identity of all life forms,
- rethinking the relationship between the living and non-living,
- the collapse of **techno-utopianism**,
- and the ethics of creating post-natural worlds.

LECTURE 3 - The Posthuman, the Anthropocene and the Post-natural in Contemporary Literature - II

One now has to speak of **posthuman** rights of **animal hybrids**, **animal-human hybrids**, **human-machine assemblages**, and new life forms and organisms modified or created by humans, including **clones** and **artificial beings**.

Post-natural wilderness

Much contemporary science fiction showcases the struggle of humanity on a toxified, eroded planet, in places where humanity is excluded or where it is too dangerous for humans to go, on the technologized earth, in a return to the land and feral lives.

Literature about **post-natural wilderness** demonstrates how humanity survives. Examples include *I Am Legend* film, Octavia Butler's *Parables* and *Genesis* series, and Atwood's *MaddAddam* Trilogy, as well as the novels of Becky Chambers, *A Psalm for the Wild-Built*.

These signal both the colonization but without deference to nature and the utilization of nature.

The **post-natural** refers to dangerous areas where humans are excluded, indicating human exclusions from these areas that make them wilderness reserves.

Although fictional strategies may differ in important ways from perhaps more narrowly defined strategies that Revlon discusses—she refers to nature writing, not fiction text in her typology—her idea of a **post-natural wilderness** set aside due to human intervention is important in understanding the novels as wilderness texts. In this sense, wilderness may also be based on land or urban wilderness.

Jeff VanderMeer's **New Weird** in *Borne* and other texts are about more than human.

Borne is a **biotech thriller** that asks what the human responsibility is towards what humans have created.

The **posthuman** is dependent on the **post-natural**, a world and its inhabitants irrevocably altered by humans.

In the post-apocalyptic world of VanderMeer, the human and non-human have all been reduced to **zoe** or their bare life. In that desolate landscape, humans do not have the upper hand, showcasing the helplessness of humans as they struggle to survive in a vision of a post-anthropocentric world.

It is not the extension of the human but rather the dethroning of the human from the center of the earth.

The end of the anthropocentric implies the end of human control over nature; the future belongs to **human-nonhuman hybrids**.

Kathy Acker's *Don Quixote, the Monsters and Freaks* and the shifting identities suggest a rejection of common humanity and point to how humanity constructs its monstrous others. In cyberpunk texts like Laura Mixon's *Glass House*, identity enables the human to process non-traumatic multiple selves, including the duality of success. The focus is on identity as not stable.

Nalo Hopkinson's *Brown Girl in the Ring* is also about unstable identities, but this time, the human identity also interferes with spirits and others.

In short, fictions such as Octavia Butler's *Speech Sounds* and Ted Chiang's *The Great Silence* blur the self-other boundary. Through their representation of the animal and the infra-human, they call for the consideration of the human and consciousness and, in the process, mobilize a **posthuman ethics**.

But nature itself is being organized, warned, and renewed by humans and technologies, so that what is natural and what is not becomes difficult to distinguish.

Examples include conservation projects, rewinding, and the resurrection of extinct species.

Rewilding

Rewilding and **post-natural** texts include fictional works like *Jurassic Park* by Michael Crichton and *Ghost Species* by James Bradley, as well as nonfiction such as *Mammoth: The Resurrection of an Ice Age Giant* by Richard Stone.

Rewilding means helping nature heal. It involves giving space back to wildlife and returning wildlife to the land as well as to the seas. **Wilding** refers to the mass recovery of the ecosystems and the life-supporting functions they provide. Rewilding means restoring and protecting specific places on land and in the ocean where nature is free to direct the ebb and flow of life. **Wilding** is about allowing natural processes to shape ecosystems so that they work in all their colorful complexity to give life to the land and the seas. Such wildlands and waters are critical to sustain ecology by supporting intact food webs and natural processes.

The idea in **rewilding** is to compensate for the **anthropogenic extinction** of several species and to repopulate the modern landscape with extinct fauna like the **mammoths**. The wilding is projected as an attempt to save the earth itself, and the return of the keystone species is not only about nostalgia for the extinct animals but a necessity directed at the future.

In **rewilding**, nature is multi-natural and technologized, thus representing a post-natural condition. It takes the laboratory into the outdoor areas and makes the outdoors a part of the laboratory.

The ethical debates around **rewilding** and the **post-natural** include questions like:

- Do extinct life forms have the same right to live as current ones?
- Should we revive them?
- Since we know little about the ecosystem for their species' characteristics, their engagement with predators and prey, and there were no humans during the age of the dinosaurs, how do we know how revived specimens will behave today?
- Is the wild species old or new?

LECTURE 4 - Posthumanism and Care

Care is traditionally defined as the work to heal and repair systems in the world. **Posthuman** approaches to care see it as a way of working towards world-making and sustaining more than human values. They reveal the role of the other material, the virus, in human life and thus alter the way we see our relationships with others.

Posthuman views of care relations recognize that **anthropocentrism** has created ethical hierarchies. There is a necessity to move beyond human care to the more than human because it attempts to acknowledge that humanity exists embedded in more than human species. That humanity and human life are dependent on a wide range of non-human species. That human life, like all life, is an emergent condition of encounters and entanglements with the non-human as well as the non-living. Who humans relate to determines who they are.

Posthuman care

Focuses on more than human effective bonds. It presents a dynamic model of care that addresses all creatures—human and more than human—as mutually constituting vulnerable entities and embedded things.

It addresses nonhumans, post-natural life, clones, and machinal care.

Posthuman care is also invested in the care of the discarded, deemed to be worthless matter. **Waste** and **discard** demand attention, thus stabilizing how we envision the norms and normative boundaries of care relations.

Posthuman care, especially in the age of robot care, asks:

- Which forms of life can give or receive care?
- If human lives are increasingly entangled with machine lives and sometimes dependent on them, and if the human family now includes robots or robotic caregivers with whom humans have an effective relationship, then does this care relation demand reciprocity, where humans care for the robot/machine?

Posthuman care asks what notions of ethics are at work in principled stances that aim to center humanity's position on earth while still stating its specific obligations.

Posthuman theories of care emphasize that human slaves have been treated as objects in history and denied care. Now objects and machines are cast as slaves, servants, and parent robots and are seen as undeserving of care. Care for the non-living, such as soil in **permacultures**, is part of several worldviews, although it has often been devalued due to anthropocentrism.

Emergence in care relations is not something that one gives and another receives; it involves entangled beings in specific situations with concrete care together.

Permaculture invites us to think with the edges of lands and systems where encounters are both challenging and diversifying beyond the expected and manageable. Embedded in the interdependence of all life forms—humans, their technologies, animals, plants, microorganisms, and elemental resources such as air and water, as well as soil we feed on—permaculture ethics is an attempt to decenter human ethical subjectivity by not considering humans as masters or even as protectors but as participants in the web of earth's living beings.

Posthuman theories of care also build on non-Western and Indigenous cosmologies that read the non-living as central to life. These cosmologies assume that the non-living, whether mineral or rock, possesses agency and even sentience. Elements like water, air, and soil must be cared for because human life emerged from these elements and is constituted by them.

Indigenous people have never forgotten that non-humans are essential because they engage in social relations that profoundly shape human life. For many Indigenous people, their non-human others may not be understood in critical Western frameworks as living. Objects and forces such as stones, thunders, or stars are known within their ontologies to be sentient and knowing persons.

WEEK 8

Human Enhancement

The moon asked, "Who wants to be enhanced?"

Everyone on Earth agreed, yet each person had a different idea of what that meant. A reclusive yogini living in the Himalayas wished to become weightless, seeking spiritual lightness. Meanwhile, an environmentalist from Costa Rica wanted to keep their beloved rainforest untouched. Most people, scarred by the last pandemic, simply wished to be immune to future viruses.

The moon questioned all beings. Most wild and undomesticated non-human animals focused only on survival, as their habitats were being destroyed. Even the "technosphere" (the interconnected system of human technologies) responded, sending signals that echoed across satellites.

Everyone and everything seemed committed to enhancing themselves. Yet, the moon felt confused. What did "enhancement" truly mean?

Is it something that can be achieved by individuals alone?

Is it a collective social project that requires regulation?

Or is it a deep-rooted drive, one that traces back to the dawn of the human species (*Homo* genus, which emerged over 2 million years ago)?

What do you think?

How would you answer these questions?

Human Enhancement: From the Individual to the Species

Imagine the possibility of **radical life extension** (extending human life well beyond its current limits) or **cryonics** (preserving bodies at extremely low temperatures, with the hope of future revival) in a world where some humans might eventually live much longer than others, or even indefinitely.

Would these long-lived individuals hold more **social and political leverage** (power or influence) than those who live shorter lives?

Would they be considered "more human" than others?

Would they dominate the job market due to their extended life spans?

Would they only form relationships or procreate with those who live similarly long lives?

Could they demand special terms in marriage contracts, such as the right to remain unfaithful

indefinitely, knowing their spouse will eventually pass away?

Would they have the right to procreate endlessly over hundreds or thousands of years on an already overpopulated planet?

Should such enhancements be considered a basic human right, or should there be limits on who gets a "second chance" at life?

Case 1: The Dictator's Paradox

Should dictators (authoritarian leaders with absolute power) be entitled to live forever?

Case 2: The Cryonics Activist Scenario

Consider survivors of extreme violence, like genocides, who choose cryonic preservation not for personal reasons, but to share their experiences with future generations.

Would their survival, based on a human rights cause, grant them special priority for future revival, so that history might not repeat the same horrors?

Existential Awareness: From Planetary to Ontological Enhancement

Can human enhancement be requested for reasons beyond personal needs and interests?

Biotechnologies (technologies that use biological systems or living organisms to develop products) of enhancement don't exist in isolation. They are part of a complex web of social, political, ecological, and philosophical relationships. **Posthumanism** (a philosophy that seeks to move beyond the traditional idea of "human" centered thinking) sees enhancement not as a purely human process but as an evolutionary change. This change affects and is affected by everything: from animals and algorithms (step-by-step procedures for solving problems), to celestial bodies like planets and moons.

As it glowed in the night sky, the moon realized something profound: everyone, everything, everywhere was already enhanced. There were no more questions, no more confusion, no more attachments. The moon, fully alight, was free—and so were we.

Moral Enhancement LECTURE 4

The Case of Joseph Jernigan and the Visible Human Project

Is there something written in our biological nature? Are humans, by definition, unlimited organisms? Do our bodies and minds need to be continuously upgraded? Where does one draw the line between bare bodies and customized bodies, between naturally embedded brains and unnaturally renovated brains? Should there be prescribed checkpoints for technological interventions in the body? Is it possible to reach our full potential, our authentic selves, and

true responsibility without the use of **Neurotechnology** or **Pre-implantation Genetic Diagnosis**?

Does the Future Belong to Frankenstein's Monsters or Moral Enhancement?

We might advocate for physical enhancement to slow the aging process, increase the height of children, repair cognitive malformations, and design future generations to be stronger and healthier. We aim to live indefinitely without finite time frames, transform and ameliorate cognitive practices, and reach a **supermind** that is one billion times more powerful than human intelligence today.

Moral Enhancement: Moral Bioenhancement and the Need for a Stable Society

Moral enhancement dreams of moral progress by claiming to:

- Control aggressive behavior
- Reduce deviant sexual behavior
- Increase responsibility towards current environmental, geopolitical, and social changes
- Eliminate vulnerability and dependency
- Prevent catastrophic climate change and ultimate harm
- Motivate action on the desire for justice
- Promote respect for authority, group loyalty, and spiritual purity
- Achieve greater happiness through the use of **psychopharmaceuticals** and brain chips
- Create morally superior post-persons who would, in turn, preserve their morally inferior counterparts

Three Options for Moral Bioenhancement:

1. Not to undergo moral bioenhancement at all.
2. To opt for voluntary moral bioenhancement.
3. To put the state in charge, making moral bioenhancement compulsory.

A Specter Haunts Our Civilization

We live in a world plagued by:

- Human trafficking
- Concentration camps
- Organized crime
- Torture and bloodshed
- Corruption, including narcissistic and antisocial personality disorders
- Defective empathy
- Xenophobia
- General unwillingness to confront the problematic aspects of the status quo

Therefore, if moral enhancement helps us prevent nuclear war, then is it worth it?

Voluntary vs. Compulsory Moral Bioenhancement

- Free will and freedom are linked to humanity, human identity, and morality.
- The “God machine” state envisions the rise of controversial moral perfectionism.
- The moral brain is deemed impossible to create.
- There is no real solution to the existence of evil, given the pluralistic nature of things.
- For example, trust and compassion benefit from the state for those undergoing moral enhancement therapies.

Designer Babies and Involuntary vs. Voluntary Moral Bioenhancement

Genome editing for the unborn includes:

- Strengthening empathy
- Attenuating violent aggression
- Improving cognitive capacities, including moral reflection

The Fantasy of Cure

Methods include:

- Omega-3 fish oil tablets
- MDMA
- Magic mushrooms
- Lithium
- SSRIs
- Transcranial magnetic stimulation
- Beta blockers
- Bioengineered oxytocin receptors
- Delivering glucose directly to the brain
- Positive eugenics for people with MAOA gene variants
- Vitamin supplements
- Brain surgery
- Elicit drugs
- Psychopharmacological treatment for depression and anxiety
- Opioid inhibitors in addiction treatment
- Nazi-level eugenic proposals
- Nanotechnology
- Human-computer interfaces
- Bionic limbs
- Automated drones

- Soldier enhancements
- Quantum computing
- Artificial intelligence
- Datafication consuming all human existence through surveillance and predictive neuroscience

The Survival Extension Debate

Post-Human vs. Trans-Human Immortalities

An **Artificial Moral Advisor (AMA)** is a form of moral artificial intelligence that could be used to improve human moral decision-making. The underlying assumption is a simple truth about humans: we are often incapable of making choices consistent with our own moral goals. Impulsive and intuitive judgment often replaces information collection, reflection, and calculation. In short, we are suboptimal information processors or moral judges.

For example, you might want to buy authentically fair trade or generally free-range products. However, when you are at the supermarket shelf, you may not have the time and mental resources to gather and process all the information necessary to align your choices with your principles. You might buy products based on their packaging, but again, your choice would often fall short of your own moral standards.

Criticism of Moral Bioenhancement (MBE)

- Morality is not something to be injected, operationally implanted, or modified.
- Morality is dynamic and context-specific; it is biological and culturally mediated.
- Freedom is non-negotiable.
- MBE could lead to automatic bio-reduction of morality, canceling personal agency.
- It may also marginalize people with disabilities and non-human others.
- The choice is between lifelong meditation, lifelong medication, or brain surgery/stimulation.
- Human nature is about openness and malleability; we need certain negative attitudes to react to injustice and violence.
- This stance is not compatible with egalitarian and liberal principles of democracy.

Post-Human Sports

We have been exploring post-human biology, post-human bodies, cyborg cultures, and bio arts.

We have further examined the hybridization of the organic and inorganic and the intersections of the corporeal and the moral.

****Human biology is not a unique, self-contained, hermetically sealed ecosystem; it is a congeries—an assemblage.**

This leads us to questions of the ethics of enhancing human bodies to think and perform better. Enhancement, as we have seen in previous lessons, is the process of transcending the limitations of the human biological form. **Transhumanists** strongly believe that the human body is a limitation, and one way to overcome such limitations is through enhancement.

Transhumanist Questions

- Why should we leave it to nature to help us overcome our limitations?
- Why should we wait for millennia when we can enhance ourselves at a faster pace to look better, develop sensitivity, perform better, think better, be physically stronger, and achieve our goals in a better manner?
- If the body is a canvas, why should we not enhance it?

Transhumanists believe that there is no need to be constrained by the limitations of anatomy, physiology, etc. The ability to evolve into something much better is something we should harness and take charge of through technology and genetic means.

Enhancing the Body as a Major Post-Human Theme

Cyborgs are one of the most popular dimensions in cyborg films. Another domain that makes this clear is post-human sports.

Athletes with prosthetic limbs compete successfully against able-bodied athletes. Oscar Pistorius, in the cult film "Blade Runner," is a reference point.

Athletes with modified limbs raise questions for the **International Association of Athletics Federations (IAAF)** regarding the use of technical devices (springs, wheels, etc.) that provide an advantage to users. Such rules essentially say humans need to remain "natural" in their competition.

The Questions Raised by These Rules:

- Are athletes enhanced humans?
- Do they possess an additional advantage due to their modifications?
- Commercialized conversions of bodies and technology already work together to enhance performance, leading to a new culture of sports.

We now see the field of **biomechanics** striving to improve performance through measurement, training, and even body modification. What about trainers and training technology with special attention to physiological and anatomical features that contribute most to participation in sports? Does only visible physical enhancement compromise performance in sports?

Commercialized Biomechanics and the Future of Sports

A "brave new sport" might follow the structure of the Olympics but with different rules based on enhanced sporting bodies.

Mental Conditioning:

- Resilience
- Staff training
- Practicing meditation
- Concentration
- Strengthening focus and determination

Are these enhancements included under the category of prosthetic devices?

When we assess a sporting body, what happens when we add chemical nutrients? Is a chemical enhancement like doping deductive? What about mental conditioning? Is being tougher a form of enhancement to compete at high levels?

What about a chess player who trains for hours and memorizes millions of possible moves? Their mind focuses and strategizes.

Related Transhumanist Questions:

Instead of training the mind through meditation and exercises for years, suppose we modified the central nervous system of a chess player, cricket champion, or football league player. Is that acceptable? Why not?

Both instances enhance abilities—why is practicing mindfulness not considered enhancement while chemical pharmacological technology is, despite both having the same consequences?

Augmented Reality and Digital Sports (Cyber Sports)

Cameras and television have altered the nature of live sports. Spectatorship has changed; we are there in the stadium. Instant slow motion, digital navigation, tracking devices, and immersive experiences have all contributed to this transformation.

Is there a body to be overcome? The question of sporting bodies cannot be segregated from the larger question of humanity and its future. It also poses questions of cost.

Research into Genetically Modifying Ourselves for the Sake of Research Quality

- Is it acceptable to augment or modify intellectual ability?
- Why are musicians genetically modified for future generations not permissible?

Gene modification is acceptable or not; education is a modification. Medical care and training facilities are always about modifying with a specific goal in mind. Talent enhances potential into a supreme quality performance.

The Conclusion of Various Transhumanist Questions

If we acknowledge the social value of enhancement (e.g., empathy), why should sports remain pure and natural?

Is there a consensus on what is "natural," and which cultures do we value over others if we may produce more, comparing ourselves to being better musicians?

Moral Enhancement

MORAL ENHANCEMENT AND CONTEMPORARY LITERATURE: OCTAVIA BUTLER'S HYPEREMPATHS

The improvement of human beings involves the condition of being human through **biomedical** and **moral enhancement**. Biomedically induced embodiment and advancement in human values, qualities, and virtues are most notably the ability to literally feel the pain of others. This serves as the foundation for a new human **intersubjective activity**: a pattern of life that subsists on understanding and responding to the pain and pleasures of those around us. It offers a way forward to counter violence by stopping dystopia.

Hyper Empathy in Octavia Butler's Parable of the Sower

If hyperempathy syndrome were more widespread, people could not commit acts of violence. They could kill if necessary, but they would feel the pain of their actions or be destroyed by it. If everyone could feel everyone else's pain, who would torture or cause anyone unnecessary suffering? I have never thought of my problem as something that might do some good before, but given the way things are, I think it would help. I wish I could give it to people. Feeling this way, I wish I could find those who share it and live among them. A biological conscience is better than no conscience at all.

The Community of Shared Pain

Medicine and drugs improve human qualities and virtues. Biomedical means of moral enhancement aim to change human nature. Civilization is on the brink of destruction; human beings need to become more human in terms of moral science, leading to morally transformed humans. This involves addressing moral operation inequalities and violence.

Communicated Pain in Pamela Sargent's "Gather Blue Roses"

I knew that she understood and felt her fear and sorrow as she realized that she and I were the same. I would experience the pain of others, feeling shattered by them. It is strange how easily communicated pain can be; I think objectively that our lives are such that comfort is often too easy.

Moral Enhancement in Richard Powers' "Generosity Enhancement"

Biotechnologically, we seek to achieve happiness and gratification through **HeLa cells**. Powers adopts a critical stance toward using biotechnology to enhance genes and thereby attain happiness. The conclusion of "Generosity" suggests that the pathway to happiness lies in hyperempathy, as presented in Butler's parables.

Moral Enhancement as a Cure for Xenophobia

- **Altruism:** A biological conscience allows for awareness of the consequences of one's actions.
- Can a human being be transformed into a being with enhanced moral values?
- Is this proposed enhancement ethical?

1. Parable of the Sower by Octavia Butler

Published: 1993

Plot/Summary: The story follows Lauren Olamina, a woman with hyperempathy, a condition that makes her feel the pain of others. As society collapses due to climate change and economic hardship, Lauren leaves her walled community in search of safety. She develops a belief system called Earthseed, which centers on the idea that "God is Change." The novel explores survival, community, and humanity's potential to evolve through understanding and empathy.

2. Gather Blue Roses by Pamela Sargent

Published: 1987

Plot/Summary: A girl named Kendra lives in a society where the weak and imperfect are discarded. She has a gift that allows her to feel the emotions and pain of others, creating a connection with those around her. The story focuses on Kendra's struggle to navigate a world while grappling with her own abilities and the moral implications of her society's values. Themes of empathy, community, and the nature of humanity are central to the narrative.

3. Generosity: An Enhancement by Richard Powers

Published: 2009

Plot/Summary: The novel centers on a girl named Thassadit Amzwar, who possesses

happiness and optimism, a state that seems to be contagious. As scientists seek to understand and replicate her condition through genetic enhancement, the story examines the ethical implications of biotechnological interventions in human emotions and experiences. The narrative raises questions about what it means to be human and the consequences of manipulating human traits. Themes of identity, morality, and the influence of biotechnology are explored.